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# Trajectory control of a laparoscopic 3-PUU parallel manipulator based on neural network in SIMSCAPE SIMULINK environment



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Neural Network;
Feed-forward;
Closed-loop Control

**Abstract** This paper presents the modeling and control process of recently developed manipulator with 3 limbs having prismatic-universal-universal (3-PUU) joints. It has 3 degrees of freedom (3-DOF), consisting of 2 rotational DOFs and 1 translational DOF (2R1T). To avoid the computational complexity of solving the manipulator's kinematics in real-time application, two artificial neural networks (ANNs) are trained to estimate the forward and inverse kinematics solutions. Training and testing results show that the developed ANNs have great prediction capabilities. The manipulator's dynamic model is deduced using MATLAB Simscape environment. Control schemes are investigated starting with motor control, first using PID controller, and then adding feed-forward control which greatly improved the motors' response. Closed-loop trajectory control based on Cartesian space feedback of the manipulators' position and orientation and inverse kinematics ANN is then studied. The closed-loop control scheme can enhance the system's performance, eliminating the error resulting from any change in the manipulator's actual model due to manufacturing or assembly defects. Simulation results of a defected manipulator model show that the closed-loop control scheme improved the manipulator's trajectory tracking capability, reducing the z-axis position error by 89.23% and the orientation error by 86.76% and 82.83% about x-axis and y-axis directions respectively.

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## 1. Introduction

Laparoscopic surgery is a type of surgery where the surgeon makes tiny incisions that are only big enough to insert special surgical instruments and a laparoscope within a trocar [1]. Laparoscopy is one of the first types of minimally invasive sur-

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gery. This type of surgeries has a great impact on reducing the patient's pain, recovery time, cost as well as the scarring in the patient [2]. However, when it comes to the surgical team, laparoscopic surgeries impose more challenges compared to conventional surgeries. These challenges include: restricted area of operation, lack of depth perception due to the two-dimensional visual feedback, reduction in sensory feedback during surgery, and difficulty of instrument manipulation due to the fulcrum effect (the instrument tip moves to the opposite direction of the surgeon's hand due to pivot point) and limited number of degrees of freedom (DOFs) [2,3]. In addition, the surgeon would lose orientation of the 3D view because of laparoscope shake if the laparoscope operator were unskilled or became tired after long operating hours [4–6]. To overcome these challenges, different robotic technologies were implemented such as laparoscope positioning robots [7–10], master–slave teleoperated systems with dedicated consoles [11–14] and hand-held laparoscopic devices [15,16].

Laparoscope positioning robots provide the surgeon with a steadier and less foggy view, eliminating the unintended moves resulting from the human assistant's hand tremor. The Automated Endoscopic System for Optimal Positioning (AESOP) developed by Computer Motion, is one of the first robots to enter the surgical arena and the first voice controlled to be approved by the Food and Drug Administration (FDA) [17]. Another FDA approved voice controlled laparocope positioner is VIKY from Endo-control Medical [18,19]. Head movement-controlled laparoscope positioners such as Free-hand ® [20,21] and the pneumatically driven EMARO [22] have also shown to be advantageous. In master–slave systems with dedicated consoles the surgeon controls the laparoscope and the surgical instruments using a master device at a separate dedicated console while monitoring the surgical view. A slave robot at the patient side maps the master's movements and manipulates the laparoscope and the instruments following the surgeon's lead [23,24]. These systems have already been used in millions of laparoscopic surgeries, mostly under the name da Vinci surgical system by Intuitive Surgical [25–28]. However, other systems might end this monopoly because of their lower cost and desirable features like haptic feedback and intuitive endoscope control [12,26,29,30]. Handheld robotic devices that integrate both the master and the slave in one handheld device combine both benefits of conventional and tele-operated laparoscopic surgeries, enhancing surgical dexterity using ergonomic handles with wrist joints at the instrument's tip and increased number of DOFs, without losing the advantages of reduced setup time and haptic feedback of conventional laparoscopy [15,16]. Fig. 1 classifies laparoscopic robots according to their type of control technique used as being open or closed loop.

Generally, laparoscopic manipulators which exist in clinical use can be classified into two main categories: serial manipulators, and parallel manipulators. Surgical serial manipulators depend on wires or rod and gear-based systems to transmit the mechanical movement to the robot joints since no actuators are allowed to be inside the patient body. Parallel manipulators are considered closed-chain mechanisms that have a movable platform connected with the fixed base platform through multiple independent kinematic chains. This configuration has offered several advantages over the serial configuration, including higher structural stiffness and payload capacity, accurate positioning with high speed, and easy sterilization process. Several designs of laparoscopic parallel manipulators have been proposed in the last decade [31–34]. Parallel manipulators that possess 2 rotational and 1 translational (2R1T) DOFs have been developed and studied such as the 3-PRS manipulator that consists of 3 limbs, each having 1 active prismatic (P) joint, 1 passive revolute (R) joint and 1 passive spherical (S) joint [35]. However, the spherical joints are troublesome and costly to fabricate. The novel laparoscopic parallel manipulator developed by Khalifa et al. [36] has 3 limbs with prismatic-universal-universal (3-PUU) architecture, also providing the manipulator with 2R1T DOFs. Unlike the 3-PRS mechanism, this manipulator does not need precise, troublesome spherical joints. Therefore, from practical point of view, 3-PUU manipulator owns better orientation capability, and is also cheaper and easier to produce [36]. The 3-PUU manipulator's performance was previously studied using PID motor control, then an adaptive fuzzy logic controller based on Fuzzy Model Reference Learning Control (FMRLC) was implemented [37]. The FMRLC controller has significantly reduced the actuators' errors with maximum error of less than 0.15 mm, 73% less than the maximum error from the PID controller.

Despite the progress of modern control theory, PID controllers are still widely used in robotic manipulators, mainly because of their simplicity and explicit tuning procedures [38]. However, additional control schemes such as feed-forward compensation would enhance performance of systems that are more complicated to control due to their coupled dynamics like parallel manipulators [39]. Feed-forward control is based on the inverse dynamics of manipulators to calculate the torques required to achieve the desired displacement at each joint and thus supply control actions before disturbance affects the process output. Recently, a proportional derivative (PD) with dynamic feed-forward compensation control system was designed for a hyper-redundant manipulator and the results demonstrated that this control method was very efficient [40]. Another kinematic redundant parallel manipulator was studied, first with PID control, then with PID plus feed-forward control. Results showed better performance after adding the feed-forward control [39].

This paper aims to numerically solve kinematics of the 3-PUU parallel manipulator using ANNs and to study the effect of several control schemes on the manipulator's trajectory tracking performance. To achieve this aim, this paper is organized as follows. Section 2 provides a brief description of the manipulator's design and presents the kinematics solution using ANNs, showing the structure and training results of the developed ANNs. Section 3 studies the dynamics of the manipulator showing detailed steps of building the dynamic model using Simscape Multi-body. This is followed by Section 4 which is dedicated for discussing the implemented control schemes and comparing simulation results. Finally, conclusion and future work are presented in Section 5.

## 2. Kinematics solution of the 3-PUU manipulator based on artificial neural networks

Fig. 2 (a) shows the schematic diagram of the 3-PUU laparoscopic manipulator. It consists of a fixed platform, a movable platform to which the surgical instrument is attached, and 3 limbs having identical PUU kinematic structures connecting

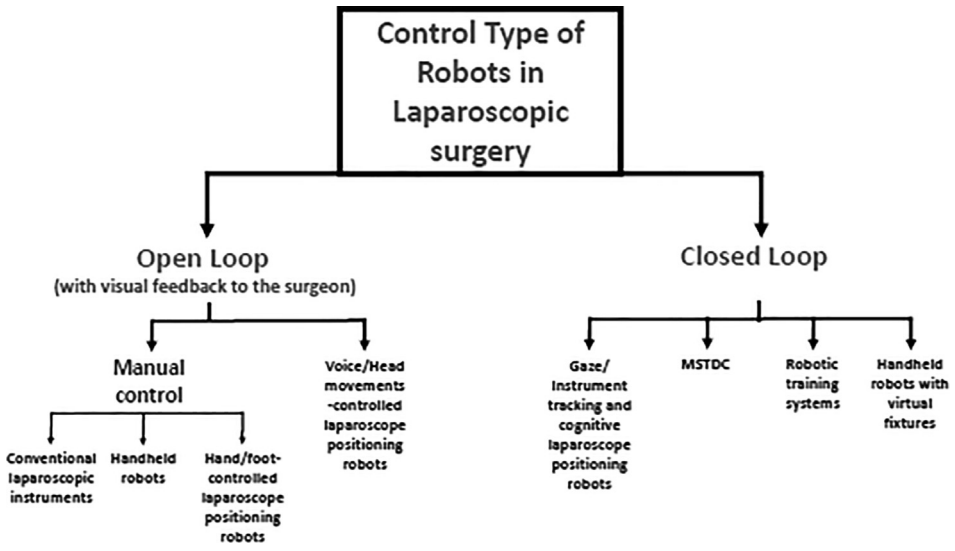



**Fig. 1** Classification of robots in laparoscopic surgery according to control type.

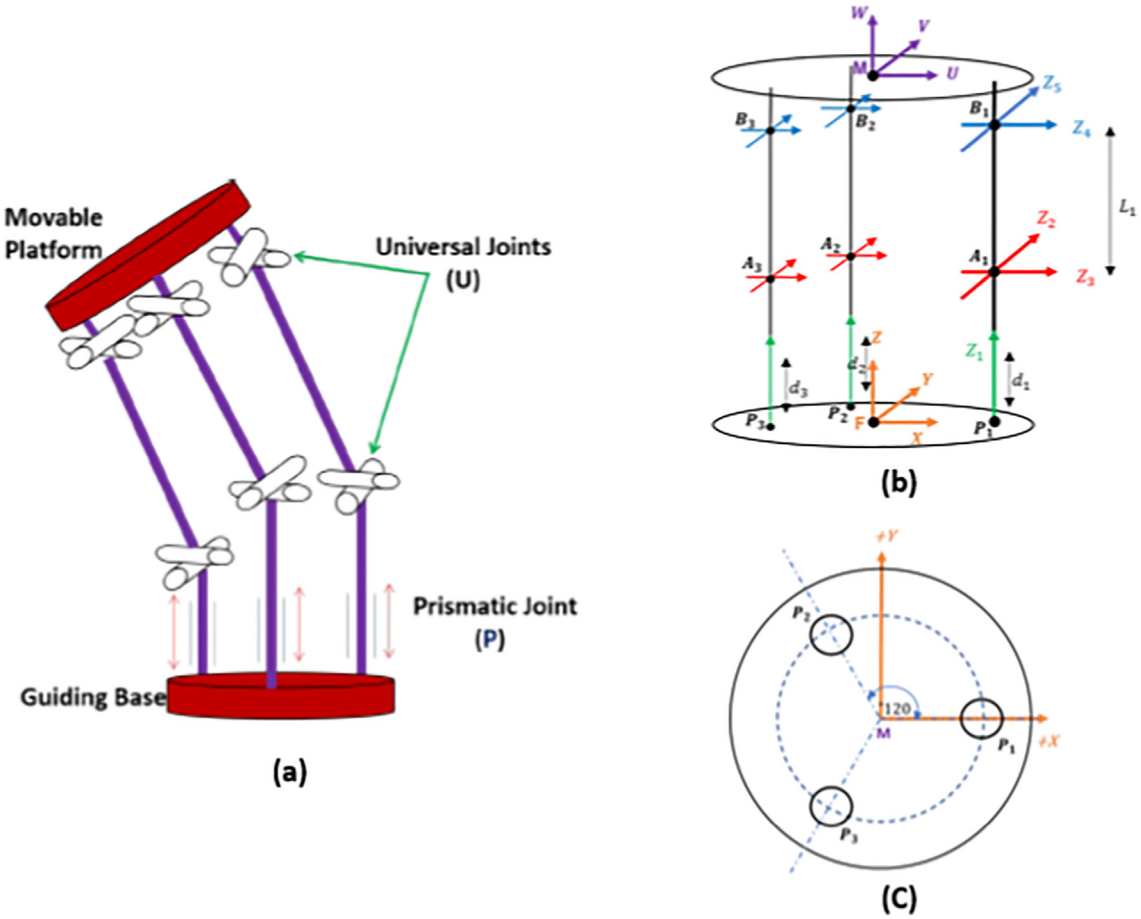



**Fig. 2** (a) Schematic diagram of the 3-PUU manipulator (b) Elevation view showing kinematics chain of the3-PUU manipulator (c) Top view of the 3-PUU manipulator.

the fixed platform with the movable one [36 41]. The first joint in each limb is the active prismatic joint that connects the fixed platform with the first link of each limb. All three prismatic joints have parallel axes that are arranged symmetrically on perimeter of a circle that lies on the fixed platform's plane perpendicular to these axes. For the universal joints, the first axis of the first universal joint in each limb is perpendicular to the plane formed by the axis of the limb's prismatic joint and the

center of the circle previously mentioned. The second axis of the first universal joint and the first axis of the second universal joint are parallel to each other and perpendicular to the line connecting the centers of two universal joints. This unique arrangement of the manipulator's joints enables unprecedented large bending angles, $\pm 90^{\circ}$ in all directions, and almost free-interior-singularity workspace [36,41]. Inverse and forward kinematics analyses are presented in the previous work of Khalifa et al. [36,41], where a closed-form solution for inverse kinematics is obtained analytically, and forward kinematics solution is obtained numerically. A brief summary of the forward and inverse kinematics is stated below.

Inverse kinematics is the problem of solving the manipulator's active joints variables ($d_1$, $d_2$, and $d_3$), given the position vector ($^{F}P_{M}$) and orientation matrix ($^{F}R_{M}$) of the end effector located at point M at the movable platform illustrated in Fig. 2. $^{F}P_{M}$ and $^{F}R_{M}$ are given according to equation (1).

$$^{F}P_{M} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}, \quad ^{F}R_{M} = \begin{bmatrix} U_x & V_x & W_x \\ U_y & V_y & W_y \\ U_z & V_z & W_z \end{bmatrix} \tag{1}$$

Inverse kinematics of the parallel manipulator is solved analytically by studying the kinematic closure for each serial chain. This results in three equations [36] (for n = 1,2,3), one for solving each prismatic displacement as demonstrated in equation (2).

$$d_n = (\,^{F}R_{M}\,\,^{M}B_n + \,^{F}P_{M} - P_n)Z - \sqrt{((\,^{F}R_{M}\,\,^{M}B_n + \,^{F}P_{M} - P_n)Z)^2 - \left((\,^{F}R_{M}\,\,^{M}B_n + \,^{F}P_{M} - P_n)^2 - L_n{}^2\right)} \tag{2}$$

Where Z is a unit vector in the z-axis direction and $L_n$ is the fixed length between each 2 consecutive universal joints in the n$^{th}$ limb. $^{M}B_n$ is a vector representing the coordinates of the second universal joint (point $B_n$) with respect to the moving base frame M−UVW. $P_n$ represents coordinates of the prismatic joint (point $P_n$) with respect to the fixed base frame F-XYZ.

Forward kinematics is the process of determining the position ($^{F}P_{M}$) and orientation ($^{F}P_{M}$) of the manipulator's end effector, given the prismatic links' displacements $d_1$, $d_2$, and $d_3$. This problem is solved numerically using MATLAB after studying the kinematic closure of each of the three limbs of the manipulator. 3 representations of the position vector $^{F}P_{Mn}$ and rotation matrix $^{F}\boldsymbol{R}_{Mn}$ (n = 1,2,3) are derived, one according to each limb [36] as shown in equations (3) to (8).

$$^{F}R_{M1} = Rot(y,\alpha_1).Rot(x,\beta_1).Rot(x,\gamma_1).Rot(y,\delta_1) \tag{3}$$

$$\overrightarrow{^{F}P_{M1}} = A_1 + Rot(y,\alpha_1).Rot(x,\beta_1).L_1.Z - \,^{F}R_{M1}.\overrightarrow{^{M}B_1} \tag{4}$$

$$^{F}R_{M2} = Rot\left(z,\frac{-\pi}{3}\right).Rot(y,\alpha_2).Rot(x,\beta_2).Rot(x,\gamma_2).Rot(y,\delta_2).Rot\left(z,\frac{\pi}{3}\right) \tag{5}$$

$$\overrightarrow{^{F}P_{M2}} = A_2 + Rot\left(z,\frac{-\pi}{3}\right).Rot(y,\alpha_2).Rot(x,\beta_2).L_2.Z - \,^{F}R_{M2}.\overrightarrow{^{M}B_2} \tag{6}$$

$$^{F}R_{M3} = Rot\left(z,\frac{\pi}{3}\right).Rot(y,\alpha_3).Rot(x,\beta_3).Rot(x,\gamma_3).Rot(y,\delta_3).Rot\left(z,\frac{-\pi}{3}\right) \tag{7}$$

$$\overrightarrow{^{F}P_{M3}} = A_3 + Rot\left(z,\frac{\pi}{3}\right).Rot(y,\alpha_3).Rot(x,\beta_3).L_3.Z - \,^{F}R_{M3}.\overrightarrow{^{M}B_3} \tag{8}$$

Where $\alpha_n$ and $\beta_n$ are the two angles of the first universal joint in the n$^{th}$ limb, while $\gamma_n$ and $\delta_n$ are the two angles of the second universal joint in the n$^{th}$ limb. $A_n$ represents coordinates of the first universal joint (point $A_n$) with respect to the fixed base frame F-XYZ. Equating the position vectors $^{F}\boldsymbol{P}_{M1,2,3}$ equations (3–5) and the rotation matrices $^{F}\boldsymbol{R}_{M1,2,3}$ equations (6–8) result in 24 non-linear equations with 12 unknowns that are the angles of the universal joints' angles. These non-linear equations are solved using MATLAB program. After solving for the universal joints' variables, these values are substituted in the previous equations to find the end effector's position $^{F}\boldsymbol{P}_{M}$ and orientation $^{F}\boldsymbol{R}_{M}$ [36]. Moreover, the manipulator's orientation represented in its 2 rotational DOFs: rotational angles about x-axis ($E_x$) and y-axis ($E_y$) can then be derived from the calculated rotational matrix according to [42].

Due to the complex geometry of the structure, as shown in equations 2–8, theoretical or algebraic methods are unable to determine the inverse and forward kinematics solution of the robotic manipulator in a fast, efficient manner. Hence, much effort has been made by researchers to develop intelligent algorithms in order to compute the inverse and forward kinematics solutions of robotic manipulators. In this work, two feed-forward ANNs with Levenberg-Marquardt back propagation training algorithm [43] have been employed to solve the forward and inverse kinematics of the 3-PUU manipulator. Supervised learning is carried out using previously generated datasets. Datasets for training, test and validation were generated using the forward and inverse kinematics solutions previously obtained.

### 2.1. Forward kinematics ANN

Forward kinematics is important in practical applications to know the exact position of the robot's end effector. Contrary

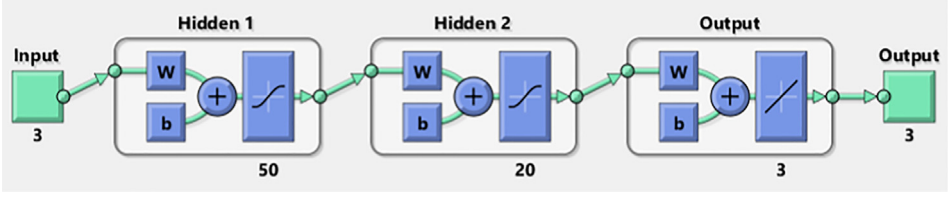



**Fig. 3** Structure of the forward kinematics estimator neural network.

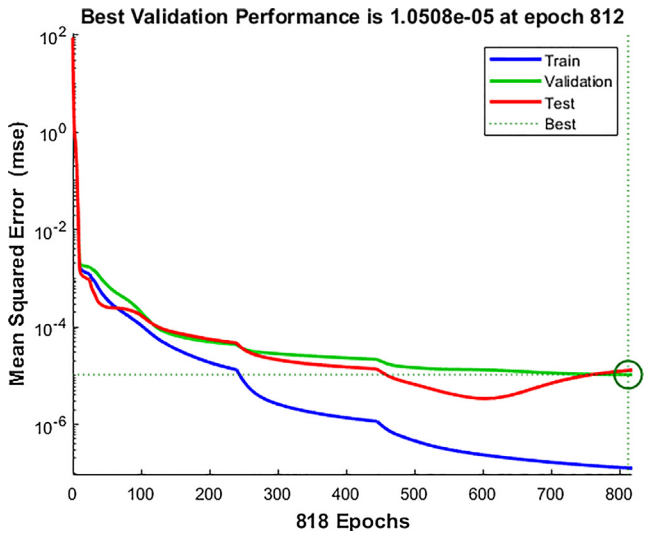



**Fig. 4** Validation performance of forward kinematics neural network.

to serial manipulators, solving forward kinematics of parallel manipulators is computationally complex and is considered a time-consuming process. This complexity is due to non-linear equations system solving. The application of ANNs in solving forward kinematics would reduce the computation time, allowing a faster, and thus more accurate real-time measurement of the manipulator's position and orientation.

The developed forward kinematics estimator ANN has 3 inputs: the manipulator's active joints' variables $d_1$,$d_2$ and $d_3$ and 3 outputs corresponding to the 3-DOFs of the manipulator: rotational angles about x-axis ($E_x$), y-axis ($E_y$), and translation along the z-axis ($P_z$). Its structure is illustrated in Fig. 3. It has 2 hidden layers (including the input layer), with 50 and 20 neurons respectively and hyperbolic tangent sigmoid function as the activation function, and the output layer with 3 neurons and a linear activation function.

This network structure resulted in best validation performance of $1.0508 \times 10^{-5}$ mean squared error at epoch 812 as illustrated in Fig. 4. The error histogram depicted in Fig. 5 shows a good normal distribution where the network has a maximum frequency of errors around zero. The regression plot in Fig. 6 shows a perfect fit between the network's output and target values indicated by an overall regression R value of 1. A comparison between the network's outputs and targets is presented in Fig. 7.

To study the prediction capability of the forward kinematics neural network, simulation is carried out with the manipulator following a certain path while measuring the simulated moving base's position and orientation using both the transform sensor blocks in Simscape and the created neural network. Fig. 8 shows the output of the neural network and the transform sensor, it is clear that the created neural network has reliable prediction capability, with maximum errors of 0.3485 mm, 0.03275 rad and 0.009409 rad in $P_z$, $E_x$ and $E_y$ respectively.

### *2.2. Inverse kinematics ANN*

Since the 3-PUU manipulator has 3 DOFs (2R1T), only 3 variables are required to describe its end effector's position and

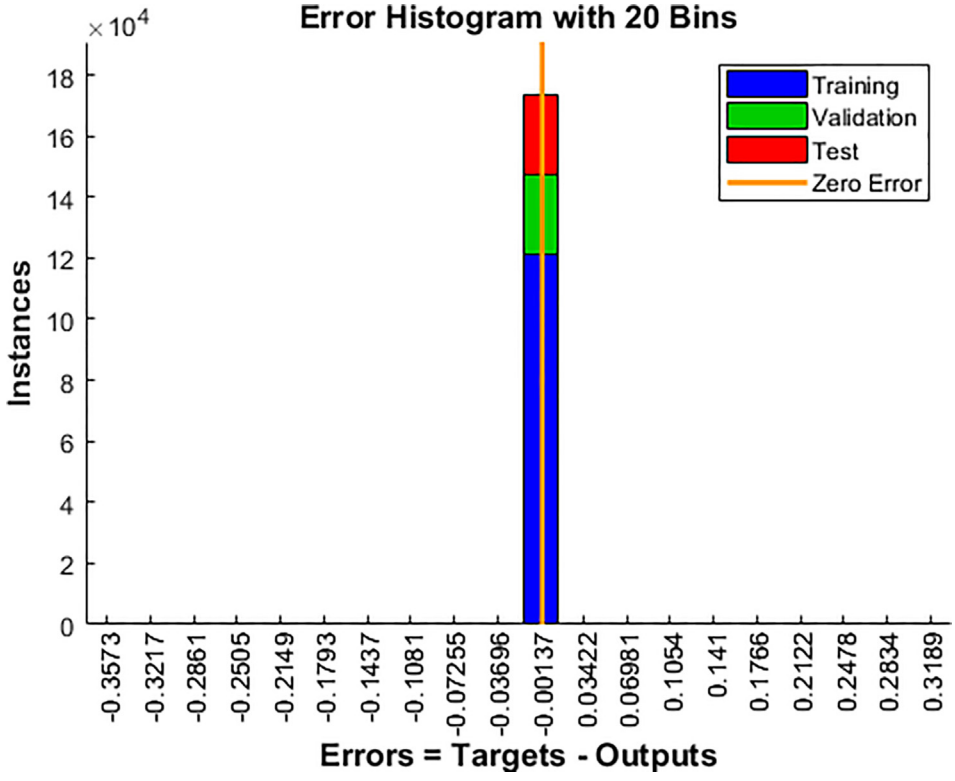



**Fig. 5** Error histogram of forward kinematics neural network.

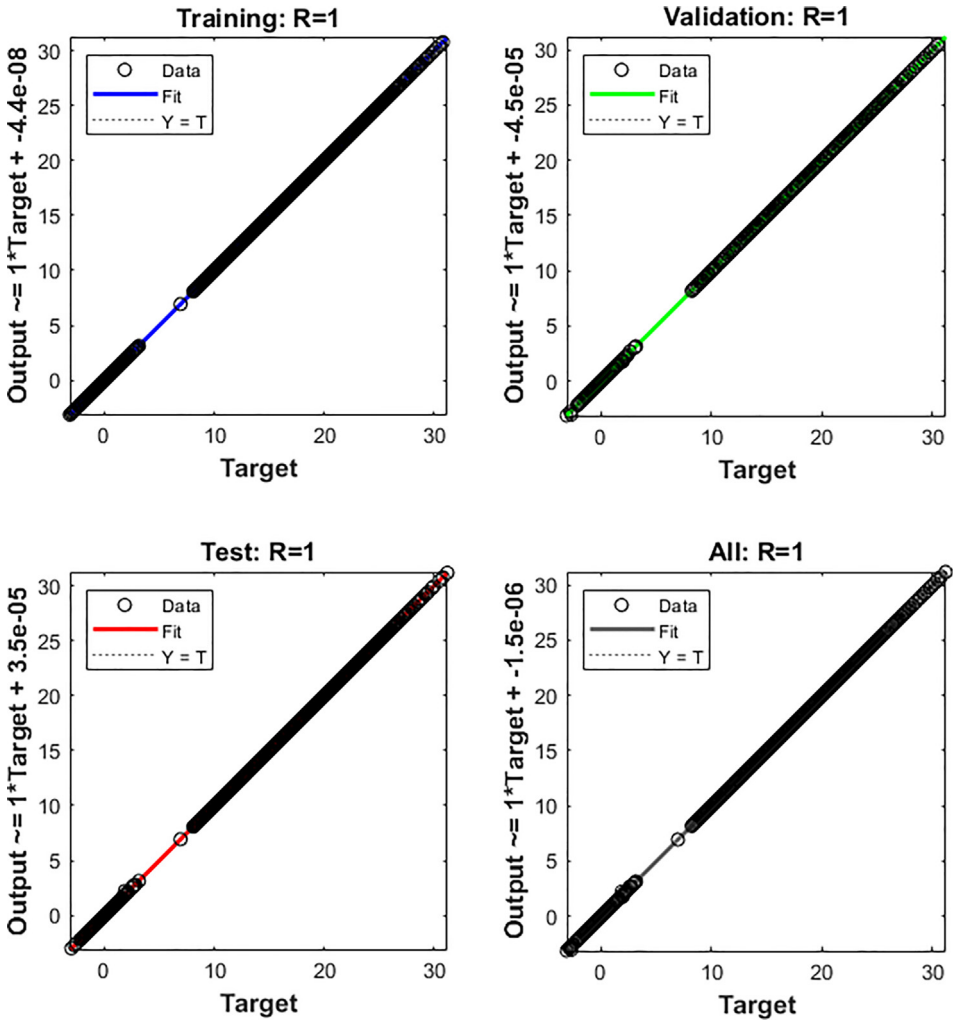



**Fig. 6** Regression plot of forward kinematics neural network.

orientation which are rotational angles about x-axis, y-axis and translation along the z-axis. Therefore, the inverse kinematics ANN is trained to have 3 inputs: $E_x, E_y$ and $P_z$.

The implemented inverse kinematics ANN has 1 hidden layer (that is the input layer) with 70 neurons and hyperbolic tangent sigmoid function as the activation function, illustrated in Fig. 9. This network structure resulted in best validation performance of $7.0226 \times 10^{-5}$ mean squared error at epoch 2119 as shown in Fig. 10. The error histogram shown in Fig. 11 shows a good normal distribution where the network has a maximum frequency of errors around zero. The regression plot in Fig. 12 shows a perfect fit between the network's output and target values indicated by an overall regression R value of 1.

A comparison between the network's outputs and targets is presented in Fig. 13.

Output of the inverse kinematics neural network is compared to that of the inverse kinematics function created using the inverse kinematics analytical solution [36] after applying new dataset as their input. Fig. 14 shows the output of the neural network and the inverse kinematics function. The created neural network has great prediction capability, with maximum deviation from the analytical solution with 0.1317, 0.2263 and 0.244 mm in $d_1$, $d_2$, and $d_3$ respectively.

## 3. Dynamics of the 3-PUU manipulator

Dynamic analysis of parallel manipulators using traditional methods like Newton-Euler and Lagrange methods would be a complicated process due to kinematic constraints of the multiple closed-loop chains [44,45]. In this work, the 3-PUU dynamic model is built using Simscape Multibody in MATLAB that provides a multi-body simulation environment for 3D mechanical systems, in addition to developing control systems and testing system-level performance without the need for any additional software.

The manipulator's links and bases are created and connected with the appropriate joints and transform blocks as shown in Fig. 15. Inertias of the bodies are internally calculated according to each link's geometry. Prismatic joints are modeled as revolute and lead-screw joints. Each prismatic link is coupled with a lead-screw joint that converts the rotational motion of the motor driven revolute joints into translational motion. Each two consecutive links are connected through uni-

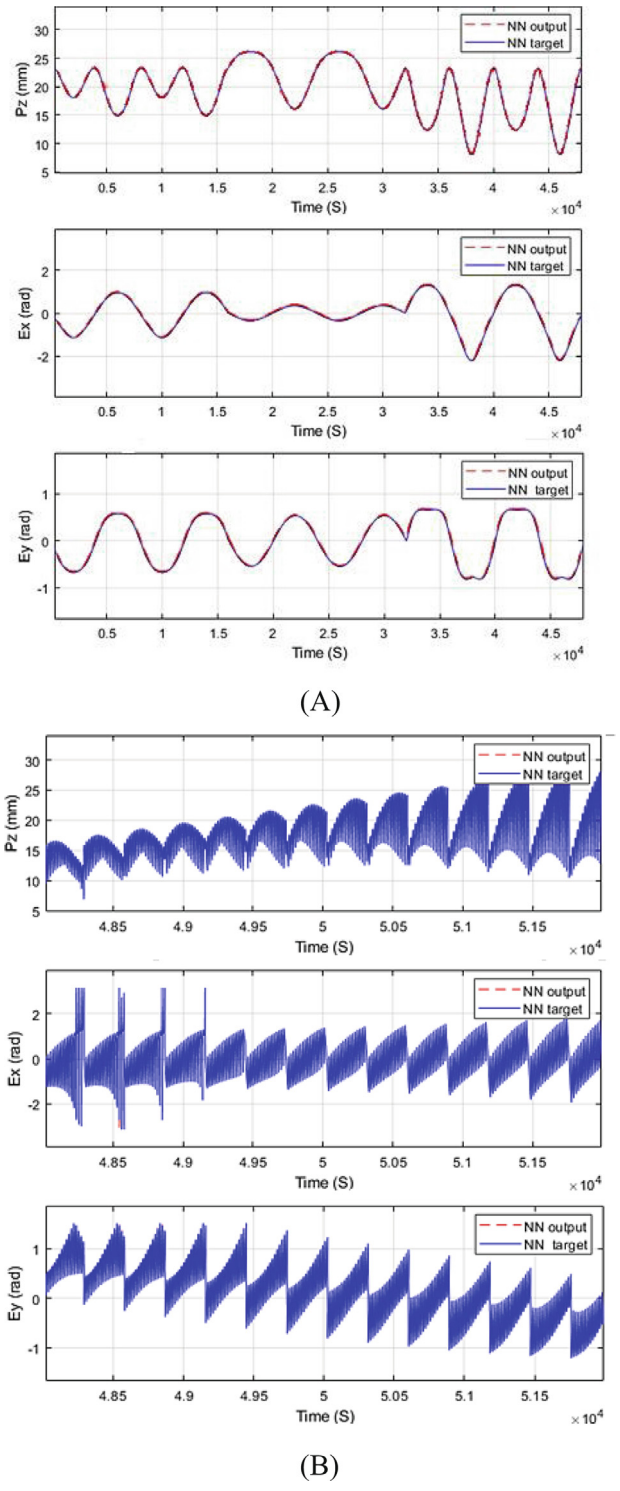



**Fig. 7** Forward kinematics neural network output versus target values. (A) First dataset (B) Second dataset.

versal joints. The virtual prototype of the 3-PUU manipulator created by Simscape is shown in Fig. 16.

The actuation method of the active rotational joints is set to be input torque and DC servomotors are added to provide the actuation torques. The used DC servomotor model shown in Fig. 17 is set to have rated torque of 8.1 Nm, rated speed of 6000 rpm, and instantaneous maximum torque of 24.5 Nm.

## 4. Control of the 3-PUU manipulator and results

### 4.1. Motor control

#### 4.1.1. PID control

Three identical PID controllers are used to control the motors' displacements, one for each limb as shown in Fig. 18. Each controller is fed with the desired prismatic displacement r(t) and the actual prismatic displacement c(t) of the corresponding prismatic link, computes the error e(t) and provides the system with the required command torque u(t) that would minimize the error according to (9).

$$u(t) = K_p.e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt} \tag{9}$$

Where $K_p$, $K_i$, and $K_d$ are proportional gain, integral gain, and derivative gain respectively that are tuned to adjust the system's response. Proportional gain reduces the error but not to zero, increasing the proportional gain introduces oscillations in the system's transient response. Integral gain eliminates the steady state error, however increasing its value makes the system oscillates. Derivative gain reduces overshoots but increasing it can make the response slower. PID parameters are tuned using trial and error method while studying the system's step response. The tuning process resulted in $K_p$, $K_i$, and $K_d$ values of 0.000005, 0.000008, and 0.0000001 respectively.

The PID controller is first tested using step input as the set point. Zero steady state error is reached after 2.4, 1.9 and 2.3 s as shown in Fig. 19 that illustrates the tracking error between the desired and the simulated actuators displacements. Moreover, to prove that the system can achieve trajectory tracking, the system is tested with applying sine wave as the set point. Fig. 20 shows that maximum error in each prismatic joint is less than 0.35 mm. Compared to the previous work in [37], this error is less than the 0.56 mm maximum error previously achieved using PID control and is larger than the maximum error achieved using the adaptive fuzzy logic controller, which was less than 0.15 mm [37].

It is more convenient to make the manipulator follow a certain path represented in its position and orientation. Therefore, the desired path of the moving base represented in its 3 DOFs ($E_x, E_y$ and $P_z$) is used to compute the required joints' variables ($d_1, d_2, d_3$) using the inverse kinematics neural network previously created as illustrated in Fig. 21. The forward kinematics neural network is used to estimate the actual position and orientation of the manipulator's moving base based on the motors' displacements.

A comparison between the desired and simulated position and orientation of the manipulator's moving base is shown in Fig. 22. It is shown from Fig. 23 that the system with PID motor control results in a maximum position error of 0.4118 mm along the z-axis and maximum orientation error of 0.05469 rad about the x-axis, and 0.03922 rad about the y-axis.

#### 4.1.2. PID plus Feed-forward control

Feed-forward control provides internal action that plans the initial motor output (motor torque) before the time when sensory feedback (measured displacement) is translated into corrective actions [46]. In other words, feed-forward supplies

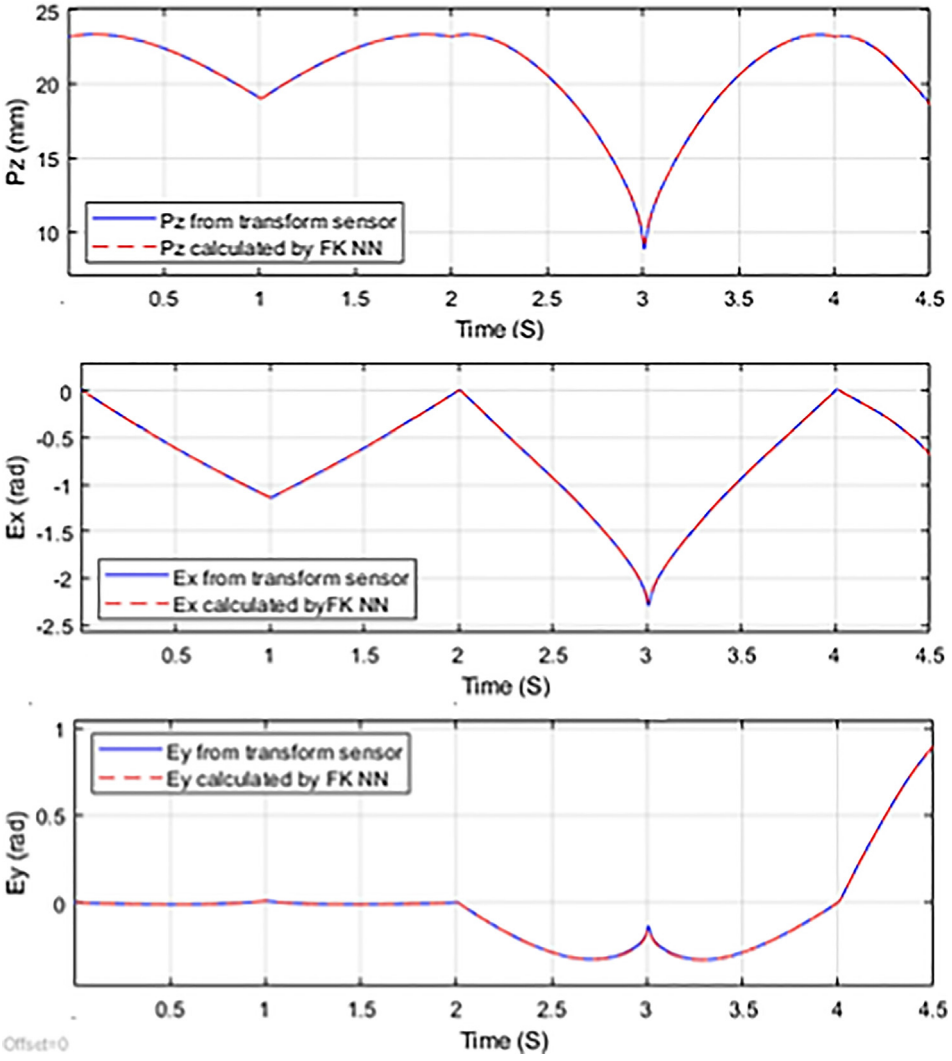



**Fig. 8** Testing the forward kinematics neural network with new dataset.

control actions before the disturbance affects the process output. This is implemented using the model's inverse dynamics that provides the torques required to achieve the desired displacement at each joint. The calculated torques from inverse-dynamics is added to the command torques resulting from PID controllers as shown in the schematic diagram in Fig. 24.

Fig. 25 shows that the maximum error between the desired and simulated displacement of each motor is reduced to 0.03261 mm. PID plus feed-forward control results in 90.2% less error than using PID control. This error is also less than the maximum error (0.15 mm) previously achieved using an adaptive fuzzy logic controller [37]. Fig. 26 depicts a comparison between the tracking error in each actuator using the PID controller and the adaptive fuzzy controller implemented in [37] along with the new PID and PID plus feed-forward control.

Moreover, Fig. 27 shows the desired and simulated position and orientation of the manipulator. It is clear that adding feed-

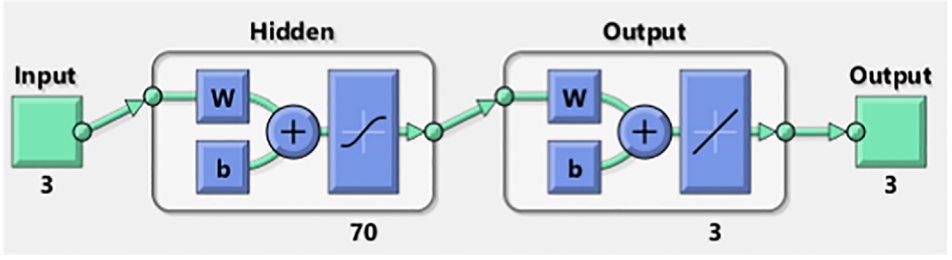



**Fig. 9** Structure of the inverse kinematics estimator neural network.

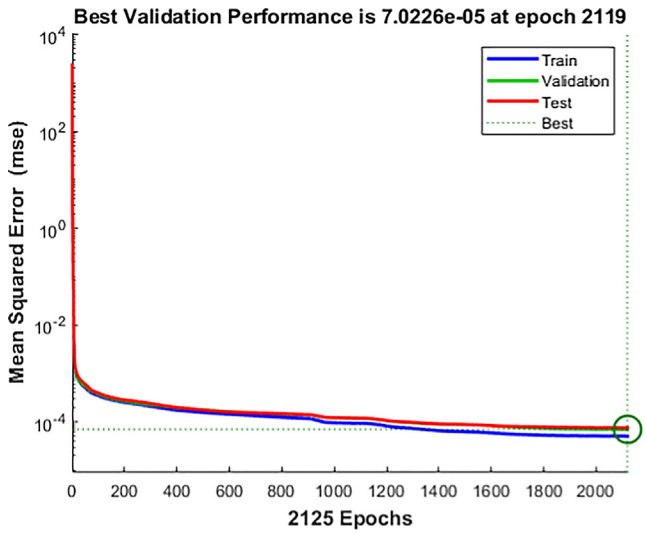



**Fig. 10** Validation performance of inverse kinematics neural network.

forward control has made a noticeable improvement in the system's response. This improvement is due to the feed-forward's principle of providing the system with a sense of the required torque before the process output (motor's displacement) is changed. Position and orientation error are shown in Fig. 28. Position error in the z-axis direction is reduced by 96.13% compared to PID only motor control and orientation error is reduced by 93.11% and 92.34% in the x-axis and y-axis directions respectively.

### *4.2. Closed-loop trajectory control*

The previous control attempts provide open-loop system performance as there is no feedback signal from the manipulator itself to the controller. They are designed to work in nominal conditions and may perform inefficiently in case of disturbances or any changed parameters in the actual manipulator due to design or assembly defects. Therefore, a closed-loop control system is required to further enhance the system's performance, and to eliminate the error resulting from any disturbances or parameter changes. This is achieved by adding feedback signals from position and orientation of the manipulator's end effector.

In this new control system, the moving base's position and orientation error is fed-back to the inverse kinematics ANN which acts as a controller. The ANN's output represents the error in the prismatic displacements $d_1$, $d_2$, and $d_3$ resulting from any disturbances or changes in the manipulator's actual model. This error is then added to the desired prismatic displacement to compensate for the actual position and orientation error as shown in Fig. 29.

To study the effect of the closed-loop trajectory control, the system's performance is tested with and without adding feedback signals from the manipulator's position and orientation. Comparison is first held without changing the manipulator's model, and then with applying changes like increased stiffness and damping of the manipulator's joints.

Comparison figures of simulation results show that there is a small difference in the system's response in case of the ideal 3-PUU manipulator with open and closed-loop control as seen in Fig. 30 and Fig. 31. Fig. 32 shows the position and orienta-

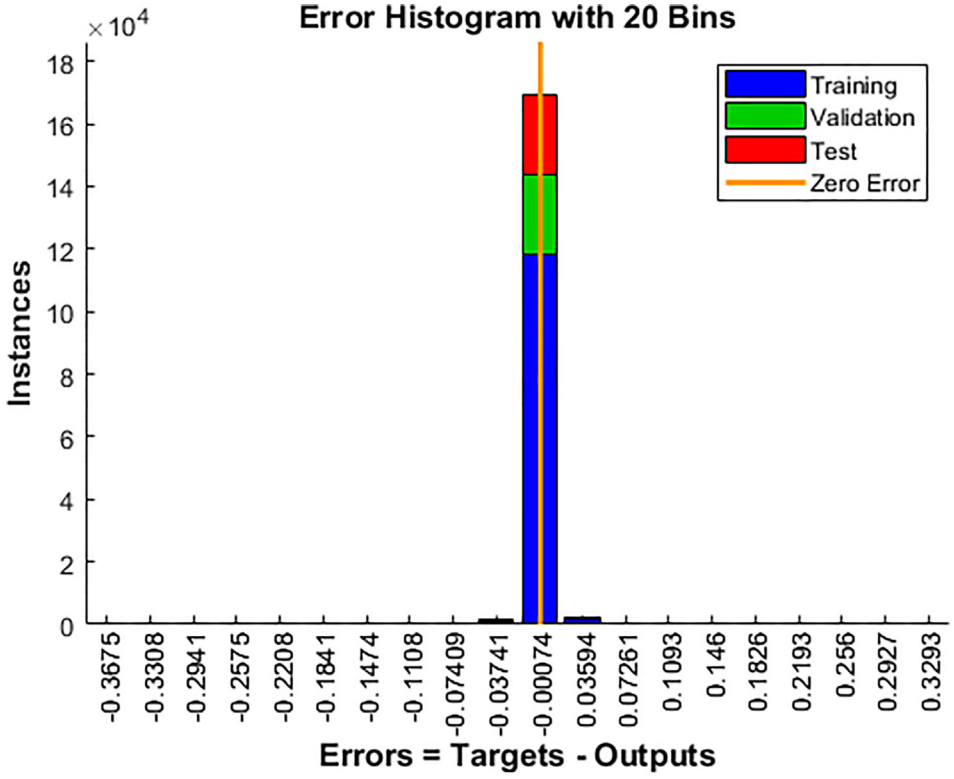



**Fig. 11** Error histogram of inverse kinematics neural network.

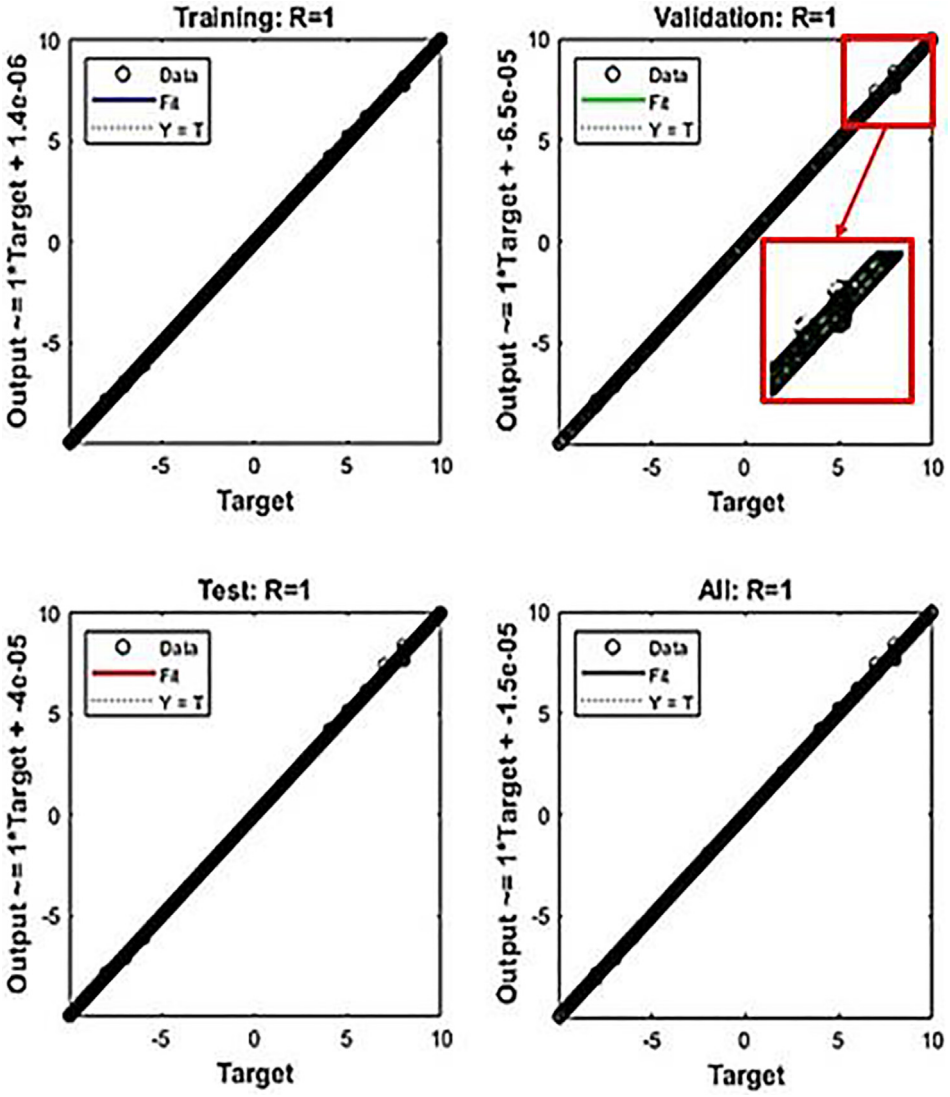



**Fig. 12** Regression plot of inverse kinematics neural network.

tion errors using open-loop and closed-loop control schemes. The maximum error in the $P_z$, $E_x$, and $E_y$ are $7*10^{-3}$ mm, $0.8*10^{-3}$ rad, $0.8*10^{-3}$ rad, respectively in open-loop control. The corresponding values in closed-loop control are $3*10^{-3}$ mm, $0.5*10^{-3}$ rad, and $0.5*10^{-3}$ rad, respectively.

Upon changing any mechanical parameter of the 3-PUU model, a case that would probably happen in real life model due to manufacturing or assembly faults, the closed-loop trajectory control can enhance the manipulator's trajectory tracking capability. Fig. 33 compares position and orientation of the simulated defected manipulator with open and closed loop control schemes and Fig. 34 compares position and orientation errors of the defected manipulator model. Closed-loop trajectory control reduced the z-axis position error by 89.23% and the orientation error by 86.76% and 82.83% about x-axis and y-axis respectively. Although the actuators' displacements might change to compensate for the error resulting from the actual manipulator's position and orientation as shown in Fig. 35.

Feedback from manipulator's position and orientation provided closed-loop control over the entire system, instead of motor closed-loop control. This results in a more robust control system with low sensitivity to disturbances compared to the previously implemented motor control schemes.

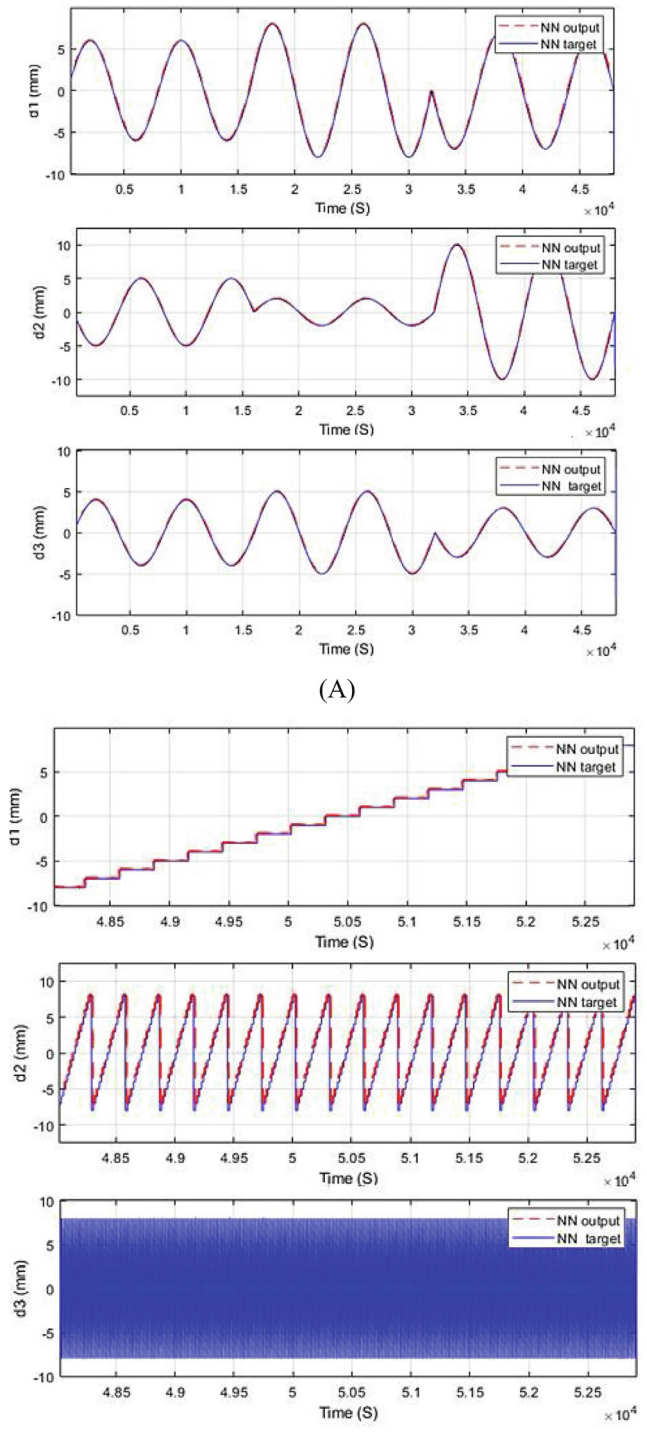



**Fig. 13** Inverse kinematics neural network output versus target values. (A) First dataset (B) Second dataset.

## 5. Discussions and conclusion

This paper presents the modeling and control process of a 3-PUU laparoscopic parallel manipulator that was recently developed by Khalifa et al. [36]. To avoid the computational complexity of solving the manipulator's kinematics in real-time application, two feed-forward ANNs were trained to estimate the forward and inverse kinematics of the manipulator. Training results showed best validation performance of $1.0508 \times 10^{-5}$ mean squared error at epoch 812 for the forward kinematics ANN, and $7.0226 \times 10^{-5}$ mean squared error at epoch 2119 for the inverse kinematics ANN. Moreover, the ANNs were tested with new datasets to study their prediction capability. Compared with the manipulator's simulated position and orientation, the forward kinematics ANN had a maximum error of 0.3485 mm, 0.03275 rad and 0.009409 rad in $P_z$, $E_x$ and $E_y$ respectively. Inverse kinematics solution obtained from the ANN showed maximum deviation from the analytical solution [36] with 0.1317, 0.2263 and 0.244 mm in $d_1$, $d_2$, and $d_3$ respectively.

The dynamic model of the 3-PUU manipulator was built using MATLAB Simscape environment, and several control schemes were investigated. Starting with motor control, PID and PID plus feed-forward control schemes were studied. Simulation results were compared to the PID and adaptive fuzzy logic control schemes previously implemented in [37]. The maximum tracking error achieved by the PID controller is 37.5% less than that achieved by the previously implemented PID control. However, it is 133.3% larger than the error achieved using the adaptive fuzzy logic controller. Feed-forward plus PID control improved the motors' response, reducing the maximum tracking error to 0.03261 mm. Compared to the adaptive fuzzy logic controller, the feed-forward plus PID control provided better motor performance with 78.26% less tracking error. In addition, this simple control scheme would be easier to implement online than the adaptive fuzzy logic controller.

Finally, closed-loop trajectory control based on Cartesian space feedback of the manipulators' position and orientation and inverse kinematics ANN was studied. The suggested closed-loop control scheme improved the system's performance, eliminating the error resulting from any change in the manipulator's actual model due to manufacturing or assembly defects. Simulation results of a defected manipulator model showed that the closed-loop control scheme reduced the z-axis position error by 89.23% and the orientation error by 86.76% and 82.83% about x-axis and y-axis directions respectively.

Future work will include experimental application of the proposed control system to validate the manipulator's trajectory tracking performance. Also, the adaptive inverse-kinematics neural network for further improvements in trajectory tracking will be investigated.

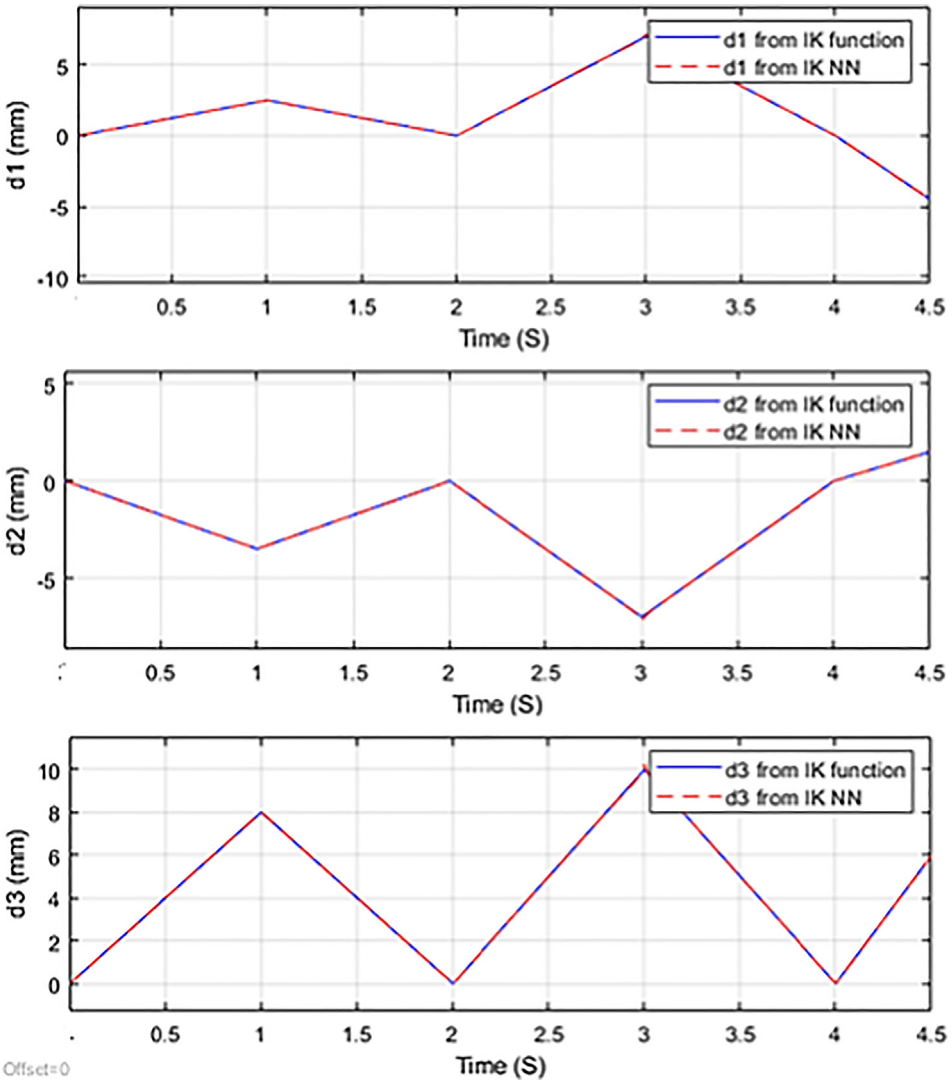
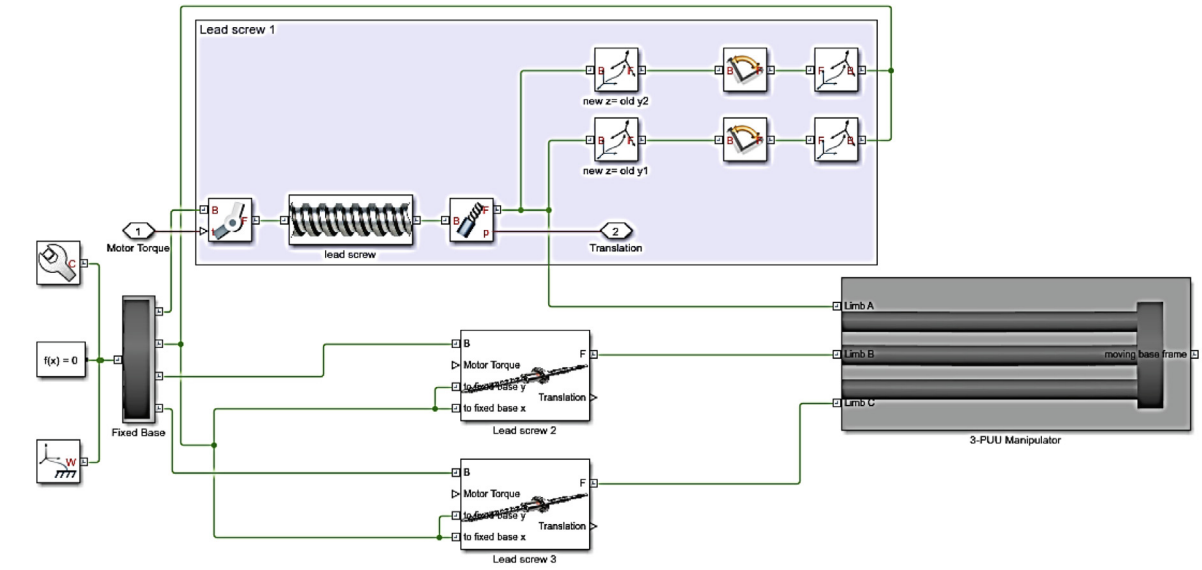
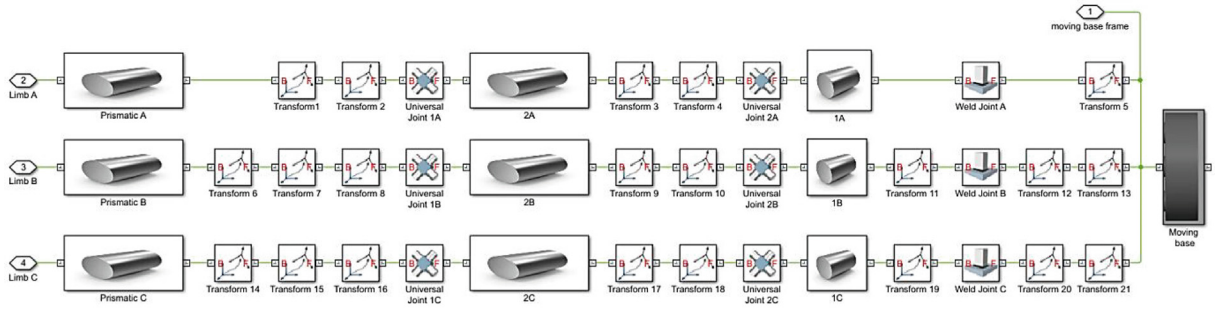



**Fig. 14** Testing the inverse kinematics neural network with new dataset.




(A)




(B)

**Fig. 15** Building the Simscape model of the 3-PUU manipulator. (A) Simscape model of the 3-PUU manipulator showing how prismatic joints are modelled as revolute plus lead screw joints. (B) Internal structure of the 3-PUU subsystem.



**Fig. 16** Virtual prototype of the 3-PUU model created by Simscape multi-body.

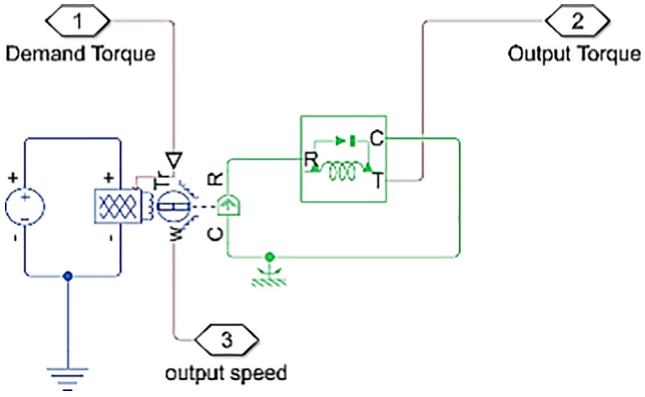



**Fig. 17** DC servomotor subsystem.

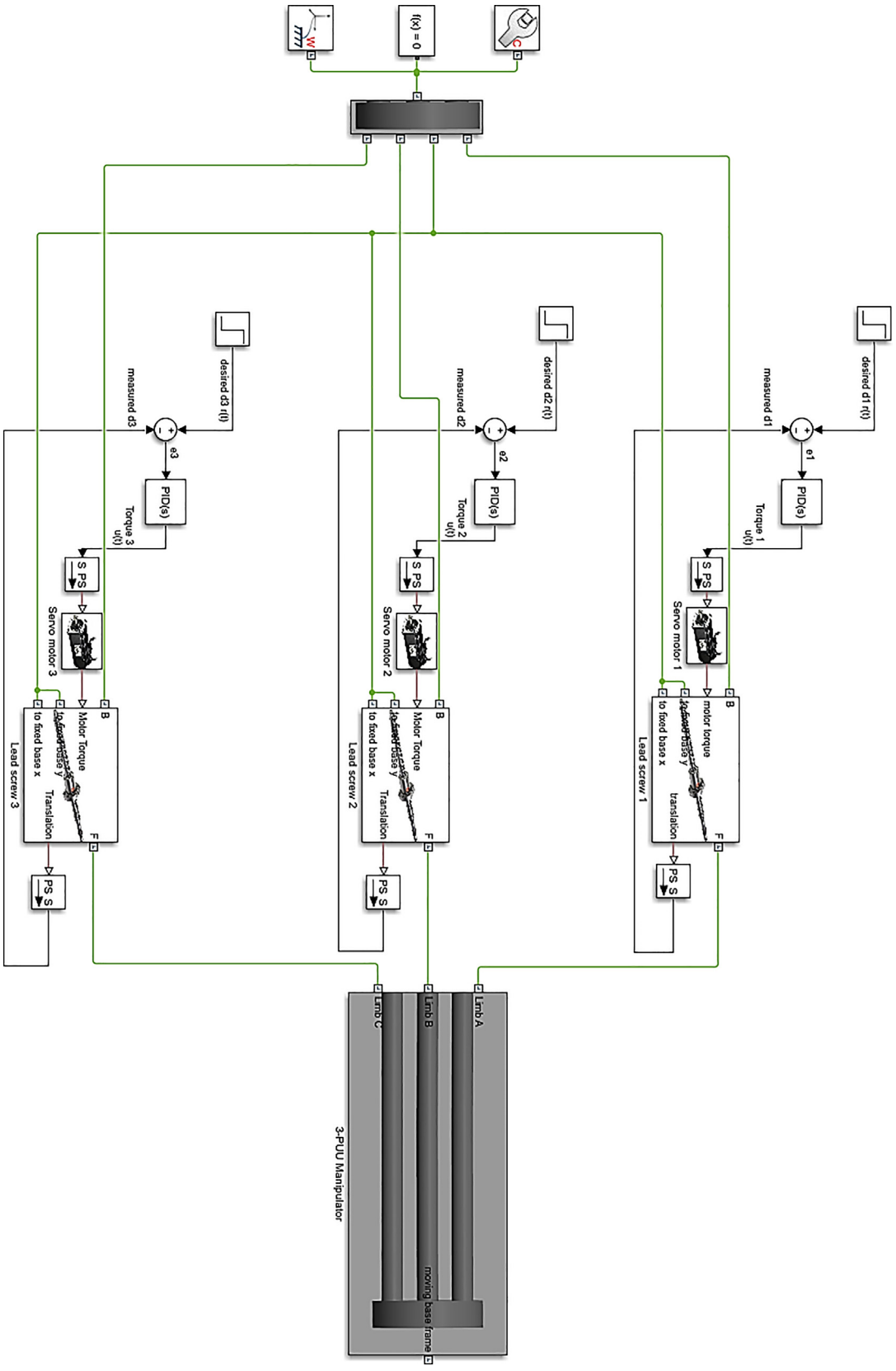



**Fig. 18** The 3-PUU model after adding the actuators and PID controllers.

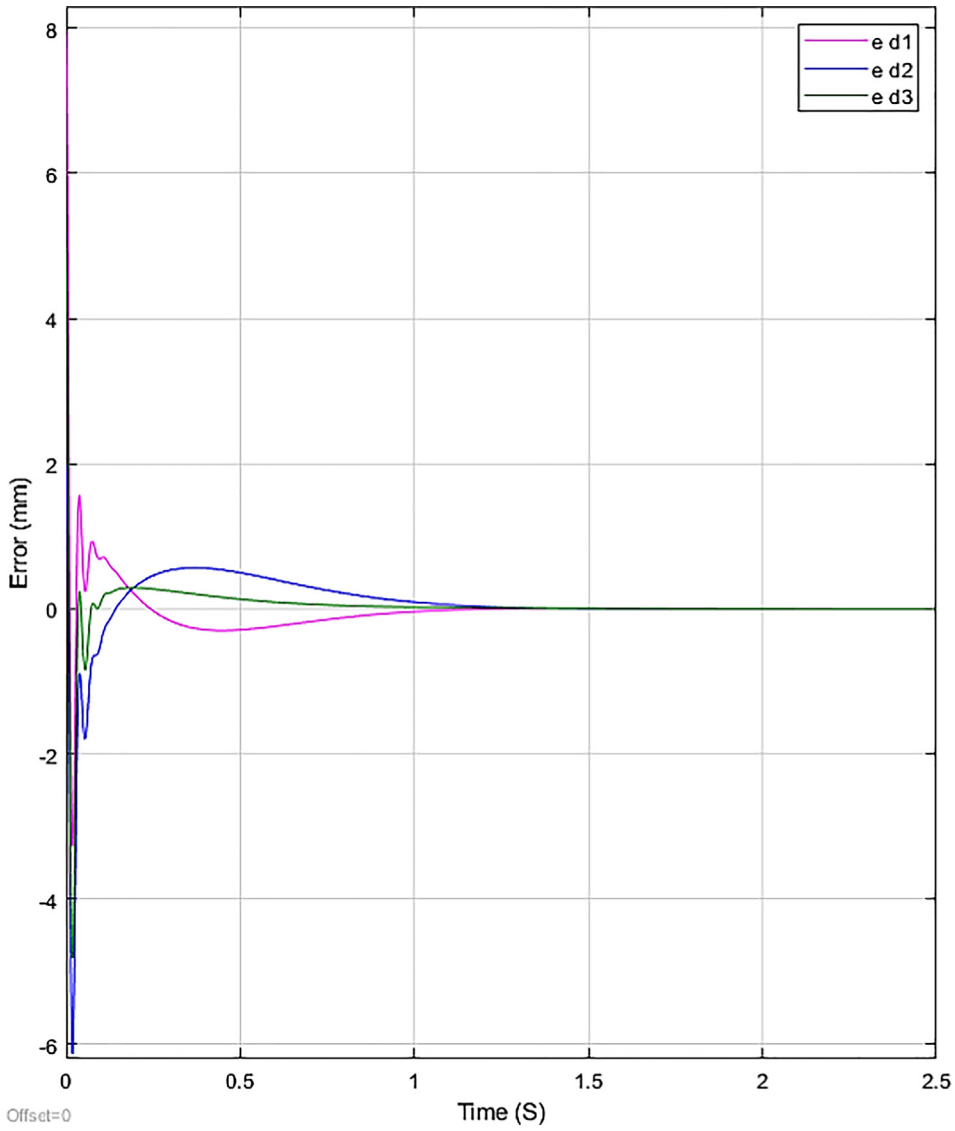



**Fig. 19** Error in motors' step response using PID controller.

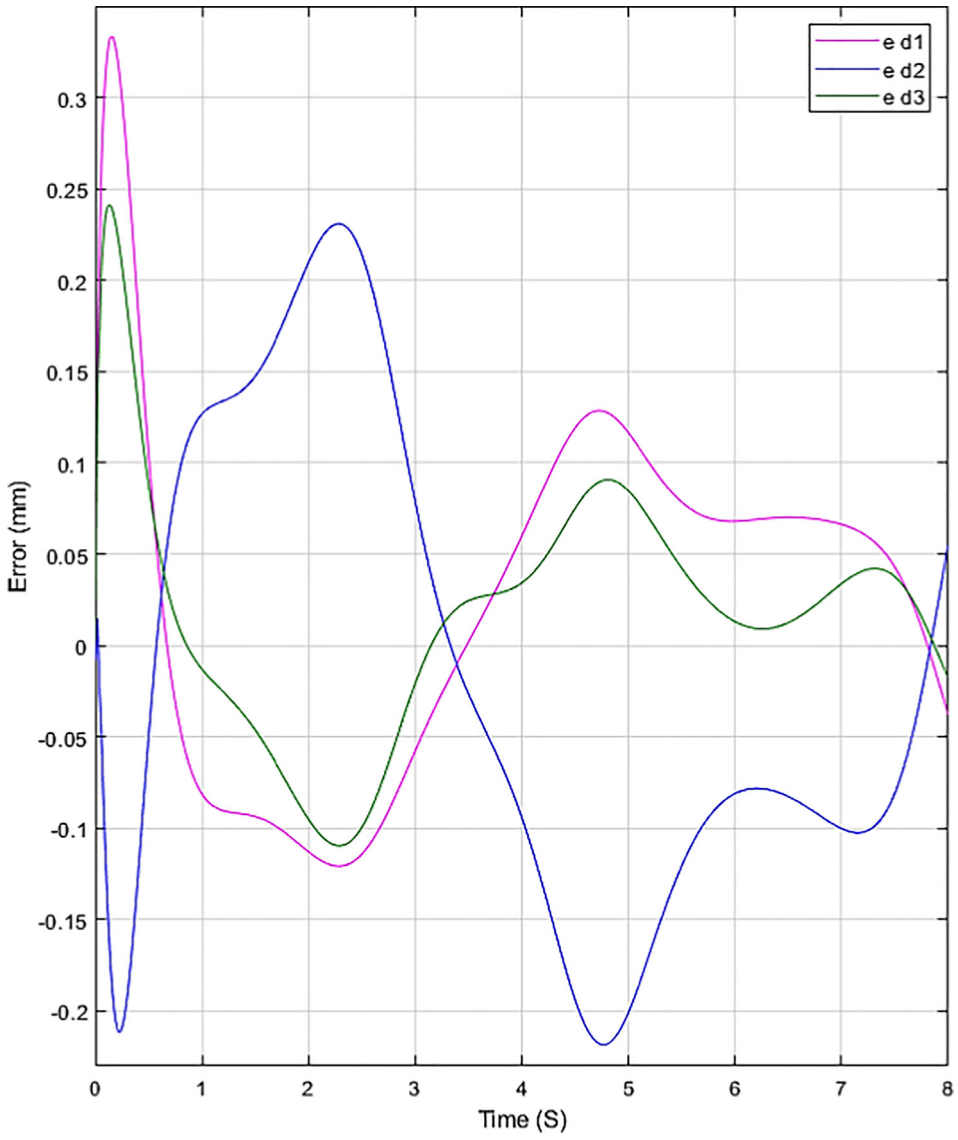



**Fig. 20** Error in motors' sine response using PID controller.

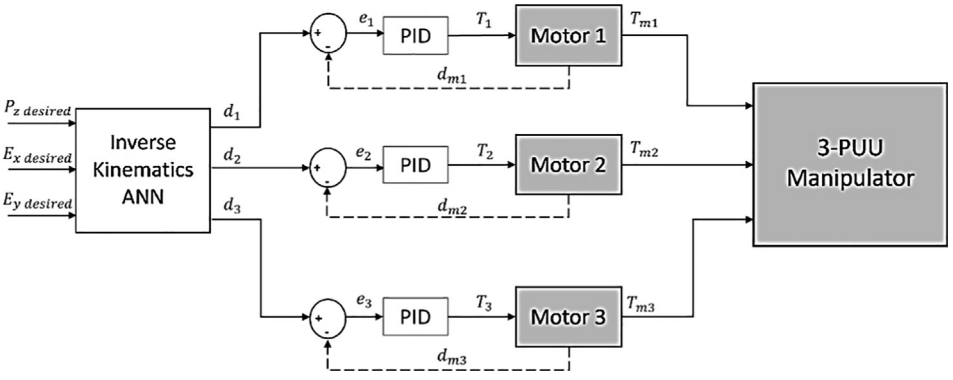



**Fig. 21** Schematic of PID motor control of the 3-PUU manipulator with the desired path as the reference input.

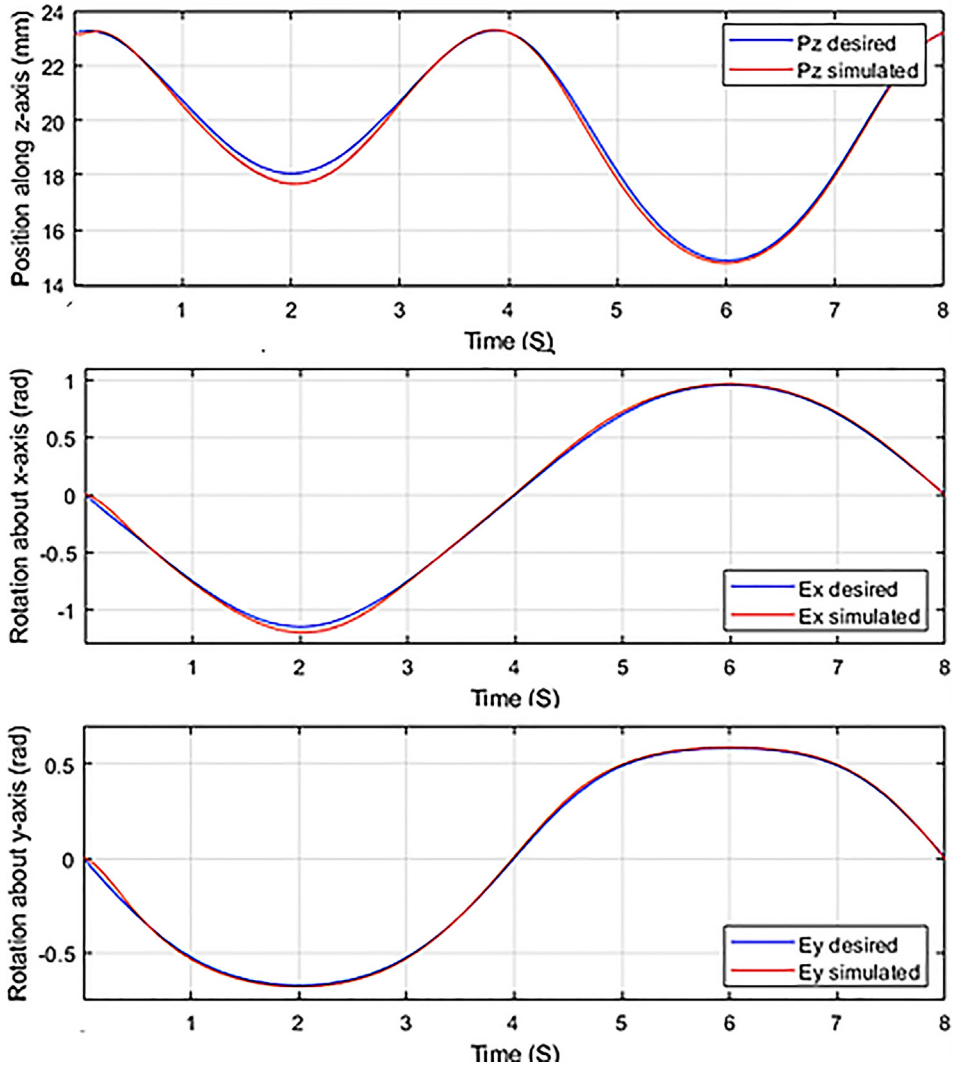



**Fig. 22** Desired and simulated path of the 3-PUU manipulator with motor PID control.

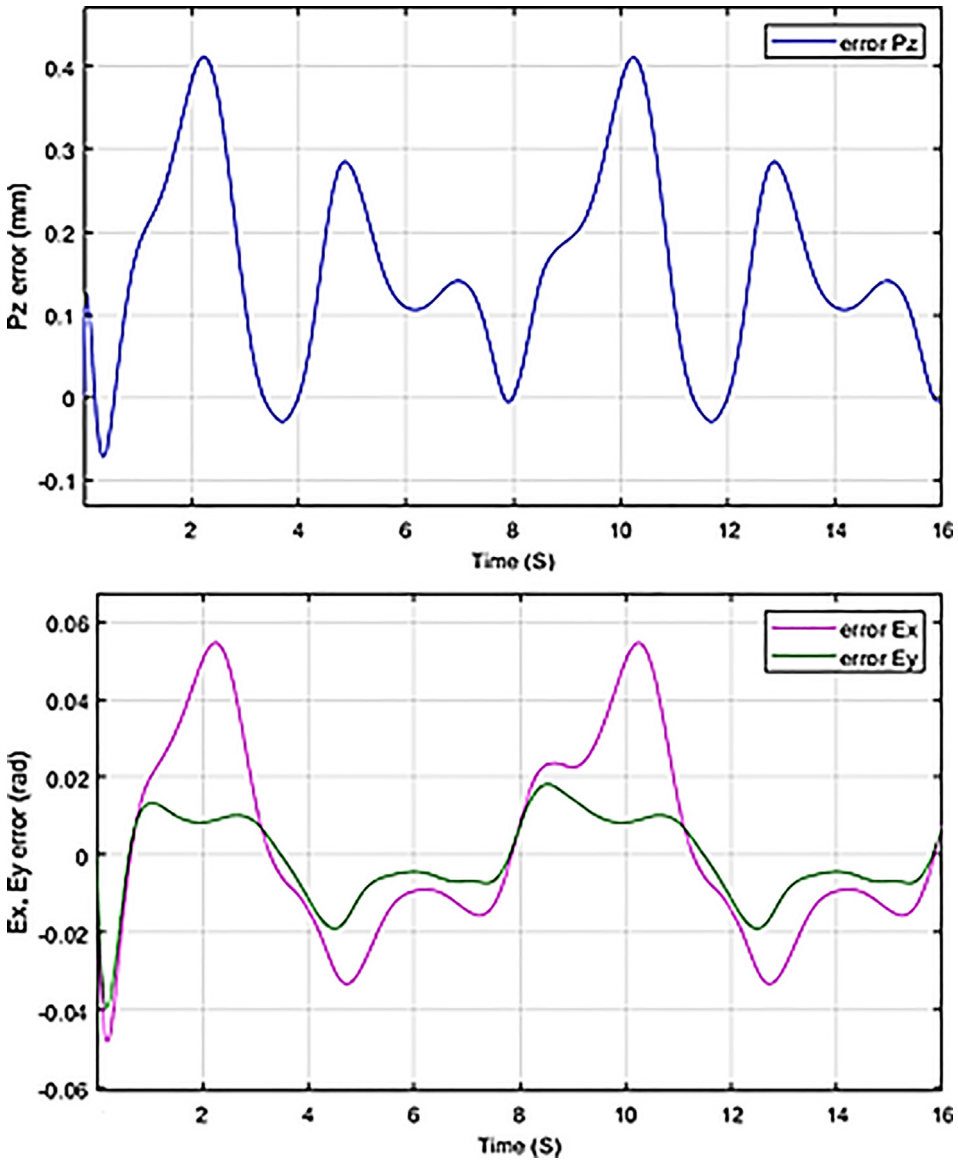



**Fig. 23** Position and orientation error with motor PID control.

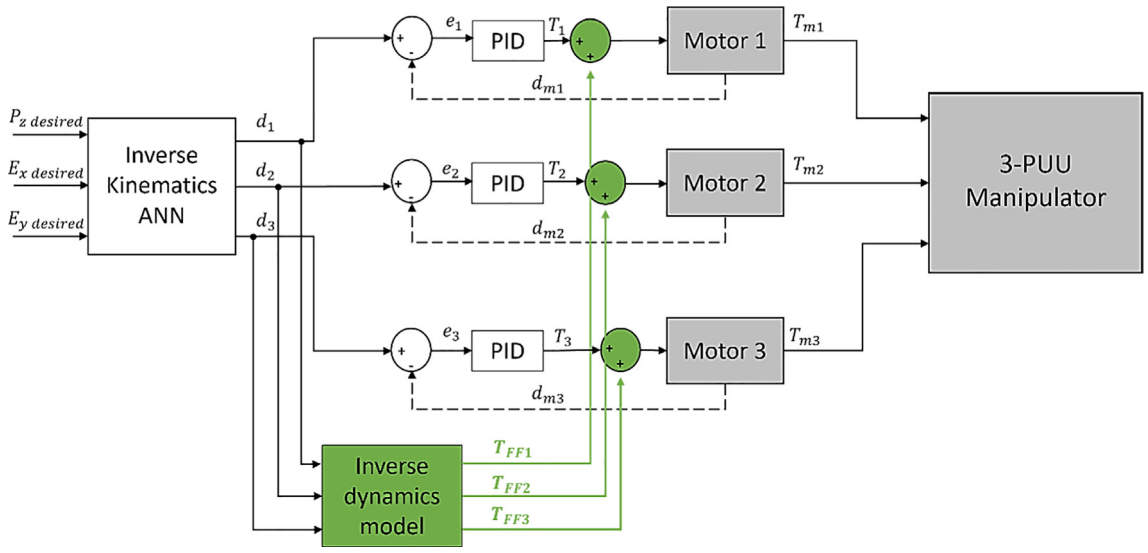



**Fig. 24** Schematic of 3-PUU model with feed-forward plus PID motor control.

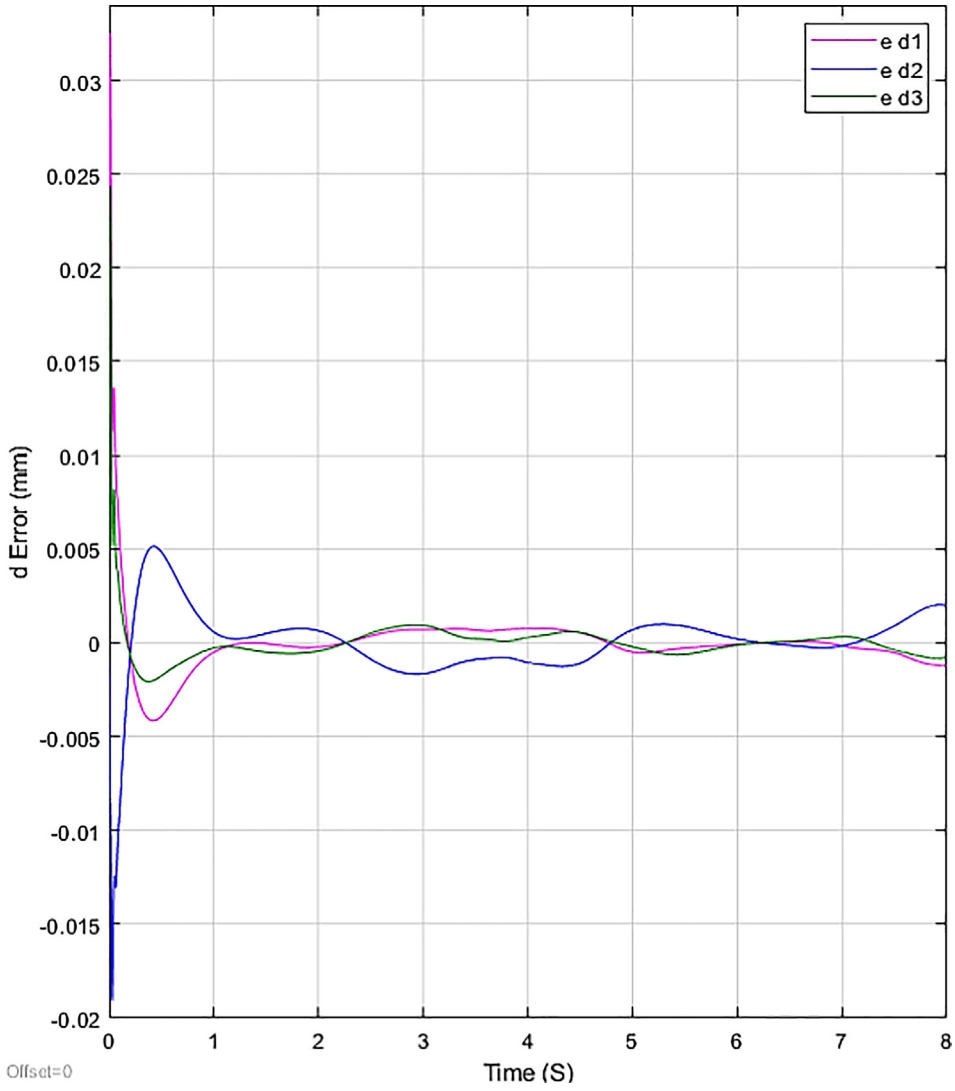
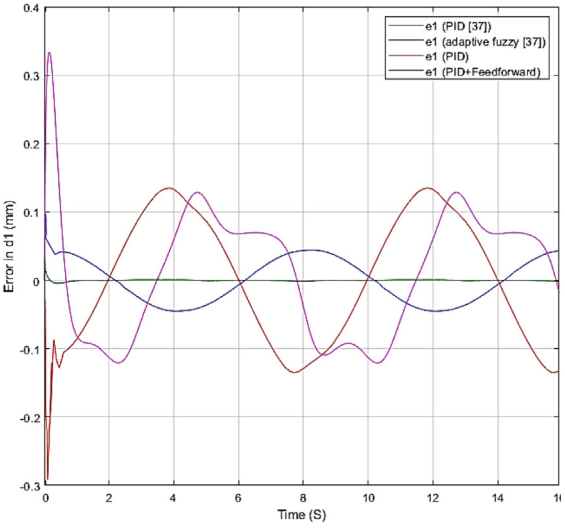
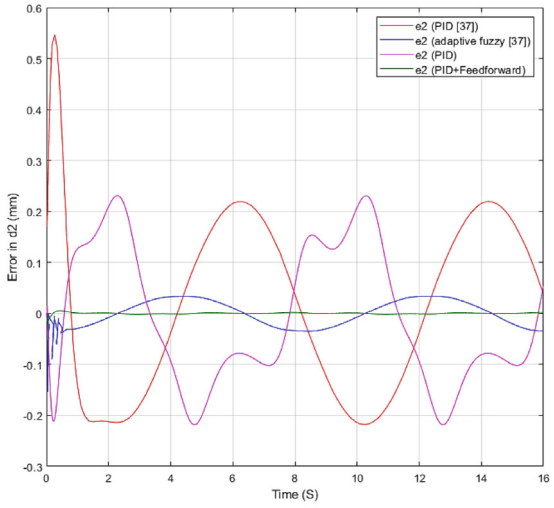


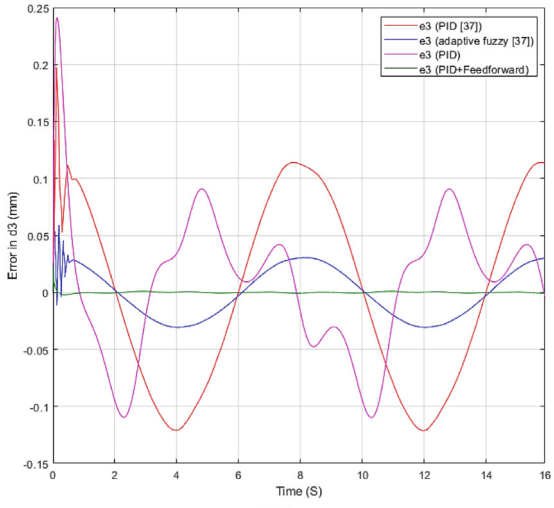

**Fig. 25** Error in motors' sine response using PID plus feed-forward control.




(A)




(B)




(C)

**Fig. 26** Comparison between the error in motors' sine response using different control schemes. (A) Error in $d_1$ (B) Error in $d_2$ (C) Error in $d_3$.

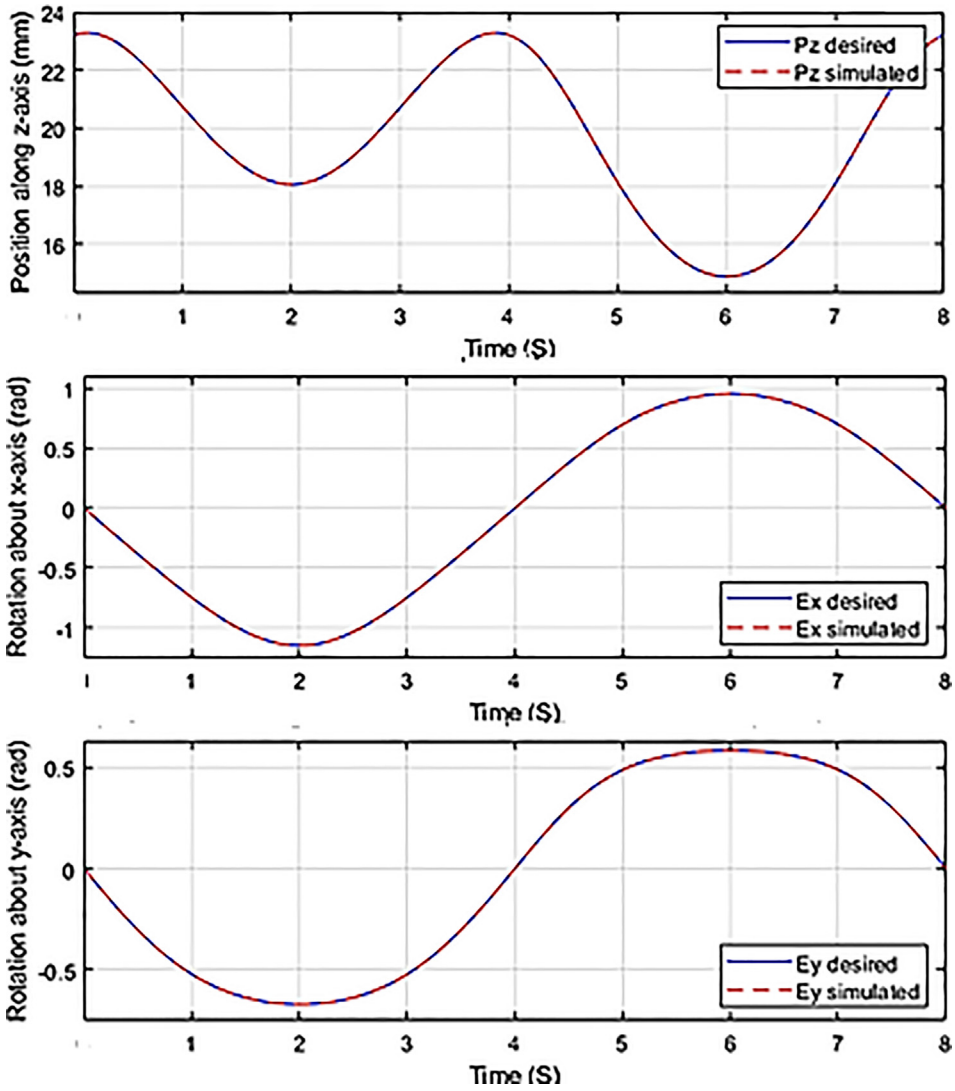



**Fig. 27** Desired and simulated path of the 3-PUU manipulator with PID plus feed-forward control.

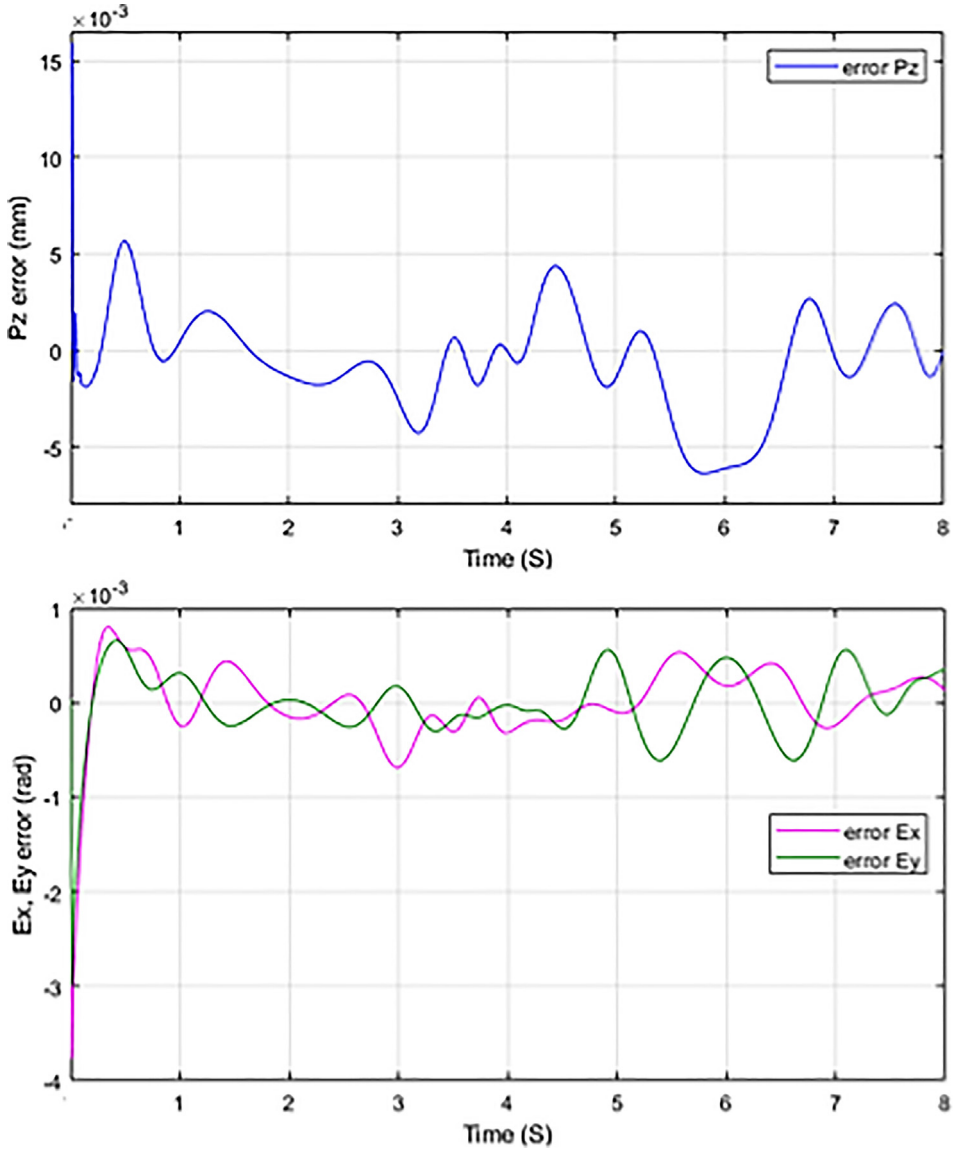



**Fig. 28** Position and orientation error with PID plus feed-forward control.

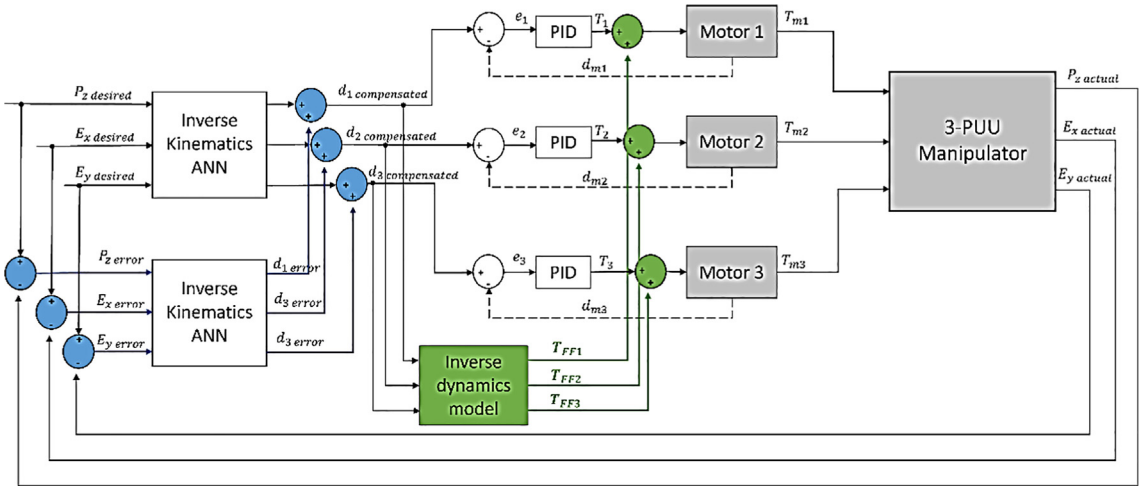



**Fig. 29** Schematic of the 3-PUU manipulator with closed-loop trajectory control.

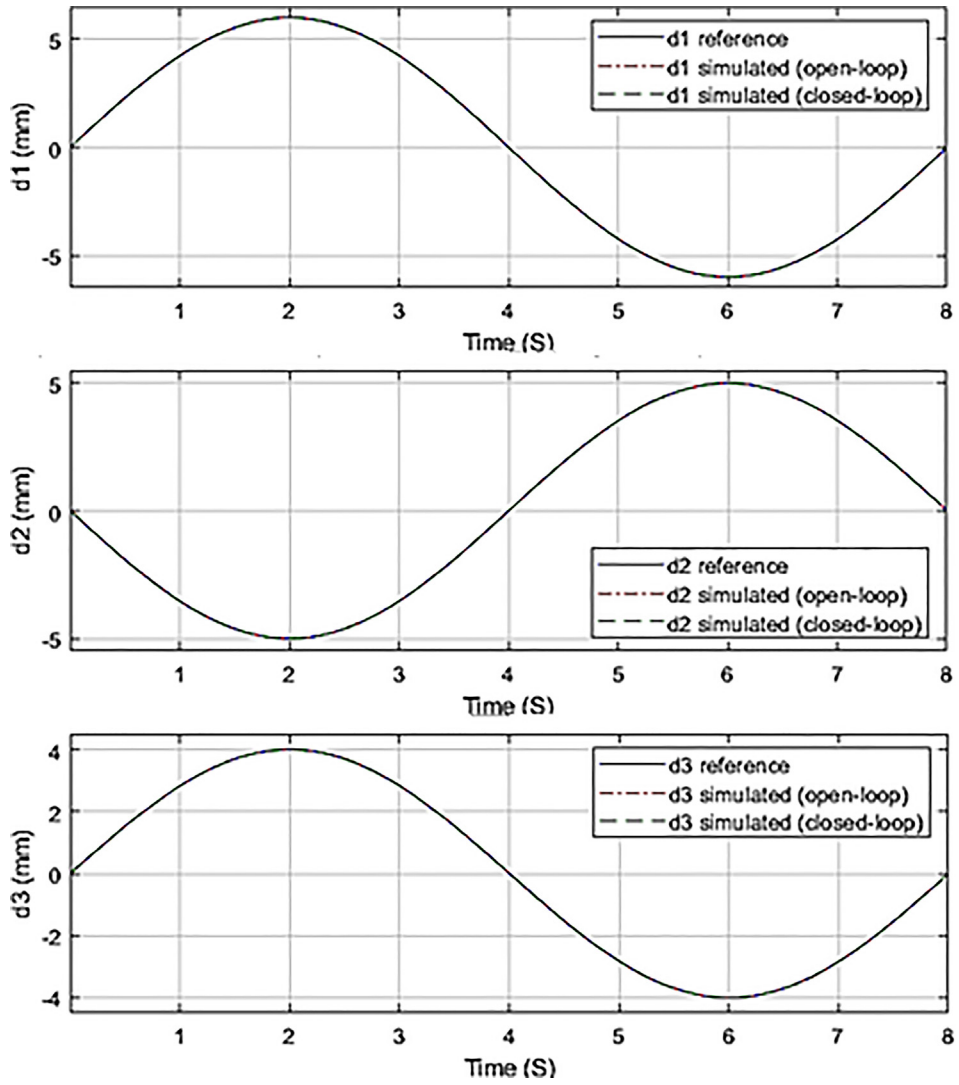



**Fig. 30** Comparison between the ideal manipulator's motor sine response with open-loop and closed-loop trajectory control.

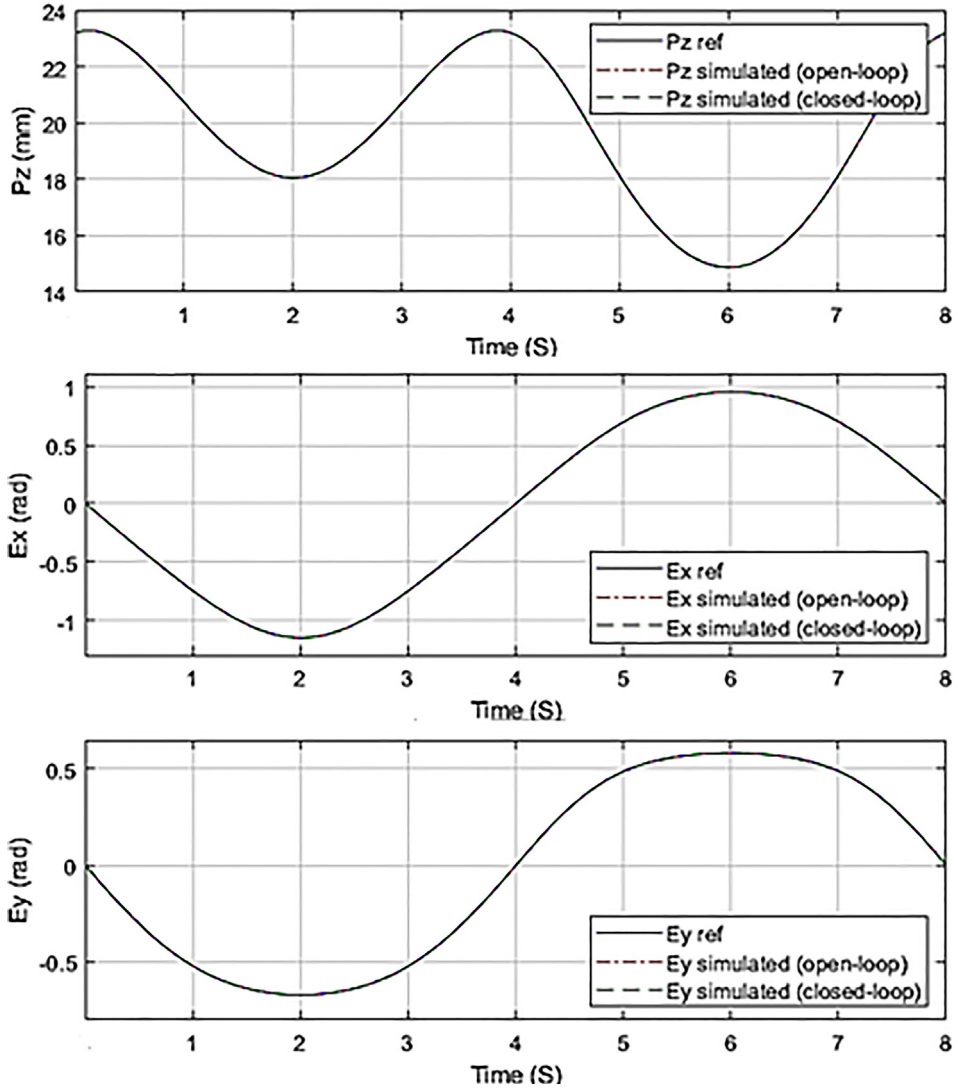



**Fig. 31** Comparison between position and orientation of the ideal 3-PUU manipulator with open-loop and closed-loop control.

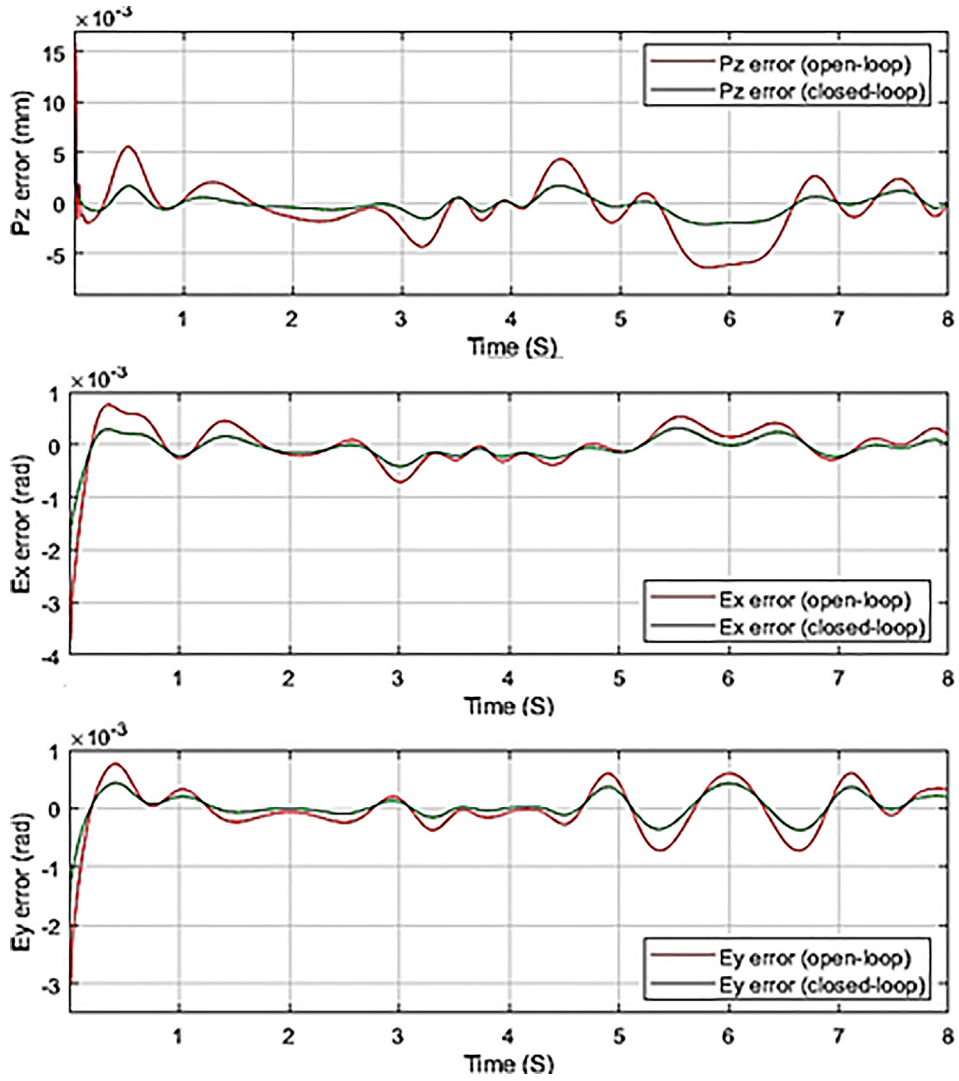



**Fig. 32** Comparison between position and orientation error of the ideal 3-PUU manipulator with open-loop and closed-loop control.

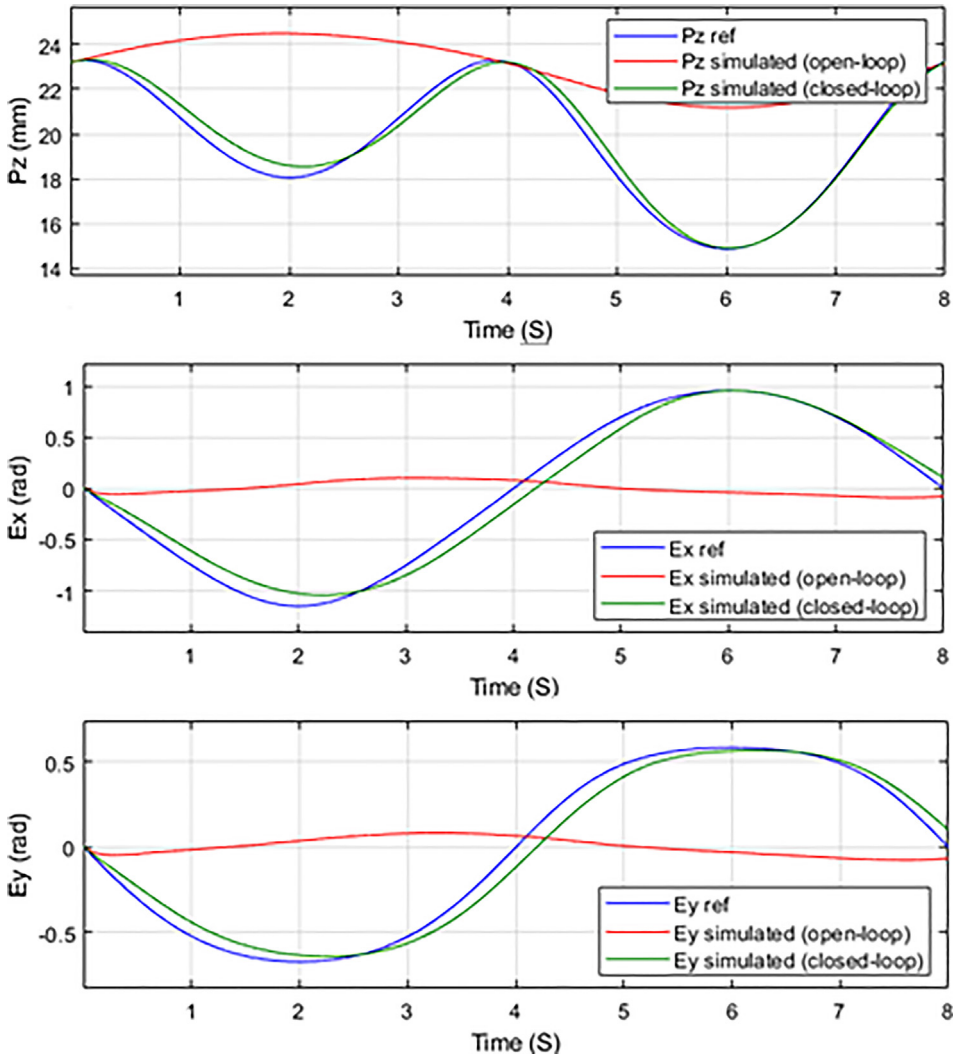



**Fig. 33** Comparison between position and orientation of the defected 3-PUU manipulator with open-loop and closed-loop control.

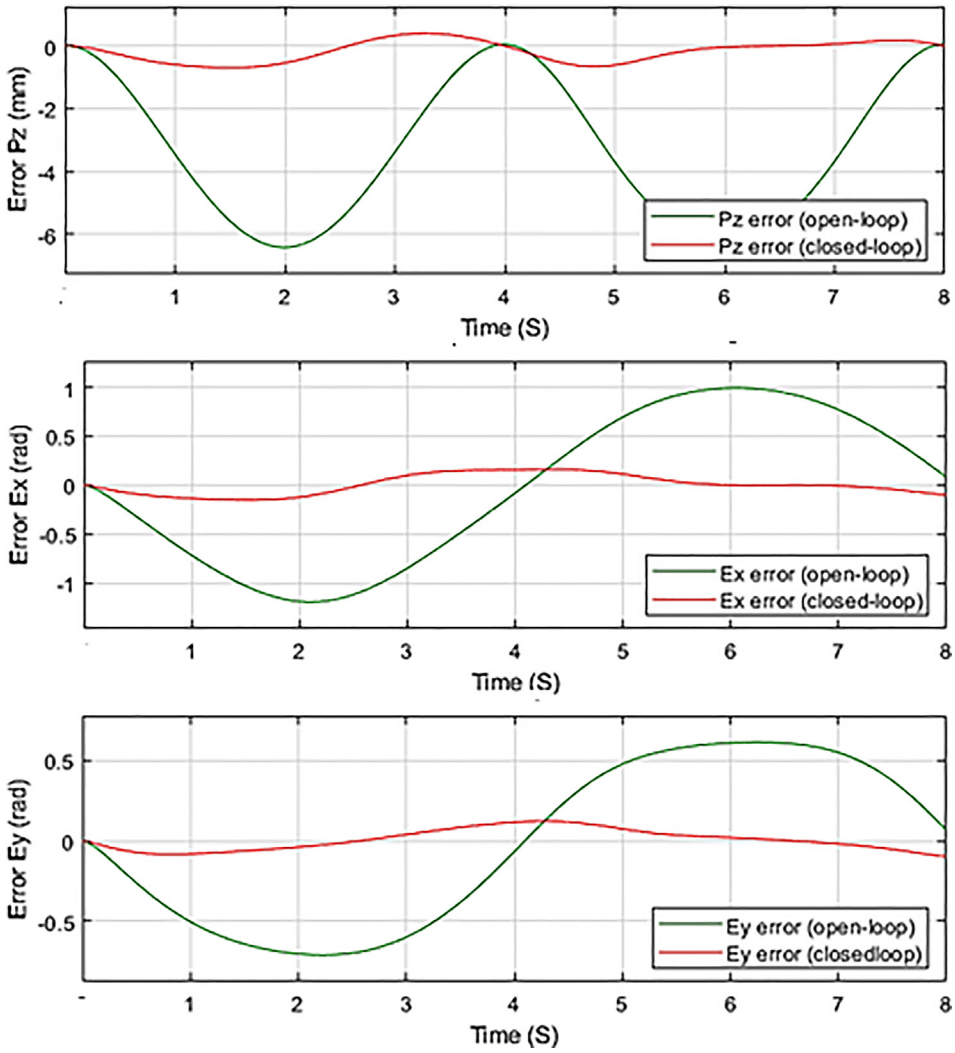



**Fig. 34** Comparison between position and orientation error of the defected 3-PUU manipulator with open-loop and closed-loop control.

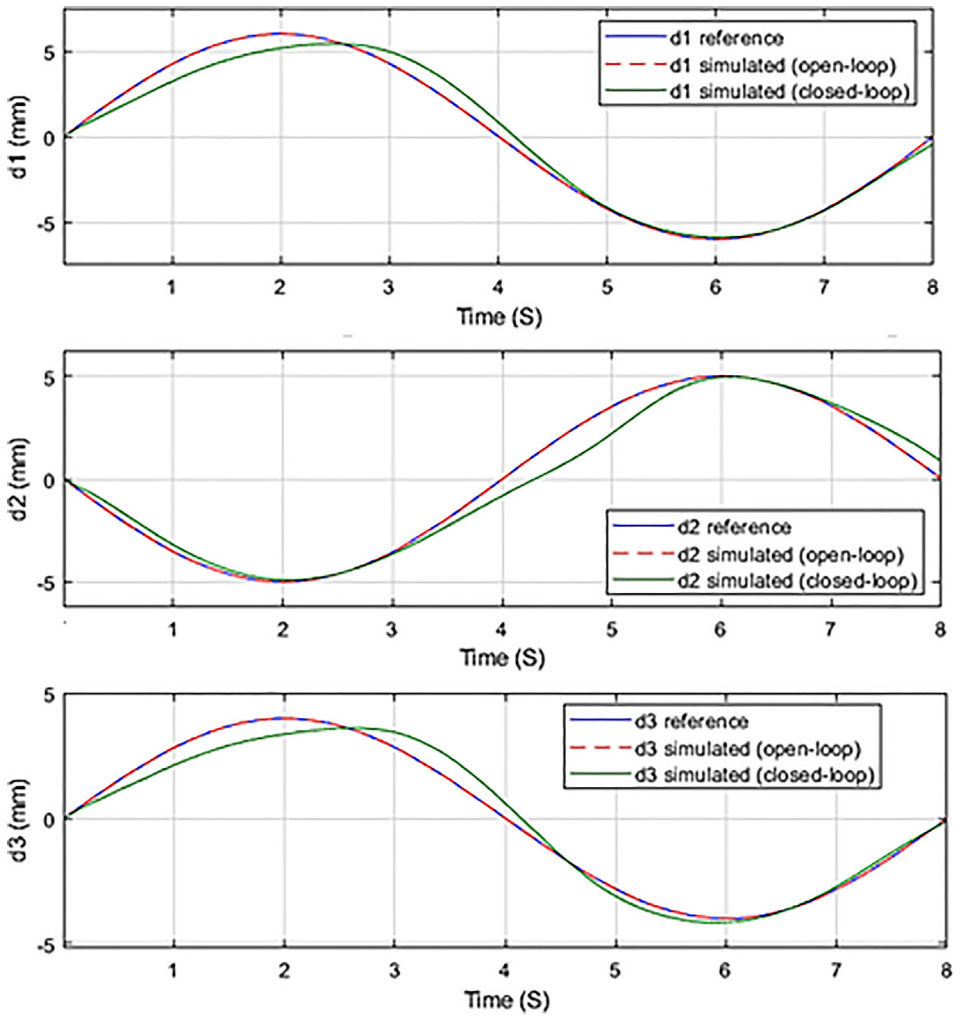



**Fig. 35** Comparison between the defected manipulator's motor sine response with open-loop and closed-loop trajectory control.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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